

Ghosts, Geometry, and Reality in Fourth-Order Quantum Theories

A guided introduction to positive metrics, Krein completions, residual cohomology, and interaction obstructions

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Abstract

Higher-derivative theories appear throughout mathematical physics and quantum gravity. Their equations contain more time derivatives than ordinary Newtonian or quantum systems, which gives them additional solutions and often improves their ultraviolet behavior. The price is the appearance of so-called ghosts: modes associated with negative energy, negative norm, or an indefinite state-space geometry. It is often said that such theories are therefore inconsistent. That conclusion is too quick: “the ghost problem” is not one problem but several — classical stability, spectral boundedness, positivity of probabilities, the choice of real observables, the construction of the vacuum, the representation of the field algebra, and the behavior of the theory when distinct modes merge into a Jordan block. Earlier work by Bender and Mannheim showed that the unequal-frequency Pais–Uhlenbeck oscillator, the simplest fourth-order system, can be converted into two positive oscillators by a nonunitary similarity transformation and quantized with a positive inner product; more recently, Bateman and Turok developed a different treatment of the resonant massless theory using a Krein space and a hidden ghost-parity symmetry. The seven technical papers introduced here do not replace those constructions. They explain their mathematical status, classify their ambiguities, identify the geometry that selects the Bender–Mannheim metric, connect the positive and Krein descriptions through a common complex vacuum covariance, and determine what changes when gauge symmetry and Lorentz covariance are imposed in fourth-order gravity. The resulting free-theory picture is that a fourth-order theory may possess one complex spectral dynamics but several inequivalent real forms and completions: the positive-metric and Krein descriptions can agree on complex correlation functions while disagreeing about which fields are real, which operators are Hermitian, and how probabilities are represented. At resonance the positive diagonalizing metric terminates, but the vacuum covariance can remain regular and continue into a Jordan theory. At the pure-Weyl boundary a global conformal deformation/BGG argument identifies Bach-flat metric perturbations modulo gauge with the parity-complete on-shell Weyl-curvature module; explicit cylinder representatives identify its D -finite $E/A/L$ realization. Paper VII then derives the certified algebraic chain from the unreduced minimal pure-Weyl BV variables $(h, c, \omega; h^*, c^*, \omega^*)$ and proves

$$\mathcal{C}_{\text{BV}}^{\text{min,alg}} \cong \mathcal{C}_{\text{raw}}^{\text{tf,min}} \oplus \mathcal{C}_{\text{tr}}, \quad H(\mathcal{C}_{\text{tr}}) = 0.$$

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The specified gauge-fixed and nonminimal sectors contract explicitly; all fifteen conformal-Killing zero modes are preserved; the dual endpoint is identified with the residual obstruction dual; the selected closed-universe BV-to-BFV suspension is normalized by $\lambda = +1$; and the post-BFV positive-frequency state ledger is complete. The centered one-particle cohomology vanishes, while exactly two weight-four vertex classes survive:

$$H_{\text{res}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2.$$

These are classical deformation classes descending to the Weyl-square and Pontryagin directions, not one-particle gravitons. The normalized $E/A/L$ module and its bosonic Fock space now admit an infinite-index energy-mode Krein completion. The residual differential has a closed maximal extension, and a bounded Cartan contraction forces its cohomology back into the same finite centered block, so the completed result is still $\mathbb{C}^2$ with Gram matrix I_2 . The theorem is quadratic and does not construct a positive graviton Hilbert space. At the reduced one-particle level, a Lorentzian tensor-curl factorization and field-action residue calculation now identify the exact Cauchy–Sobolev space and prove its Krein-unitary equivalence with the energy-mode completion. An exact ghost biwave, an ordinary-derivative four-row symbol witness, and an exact 66-to-30 Fourier-complex retract with support-local formulas are now proved. The curved lower-order witness/retract and adjoint table, causal Green homotopies, the complete covariant/Cauchy pairing comparison, distributional/Hadamard completion, nonlinear stability, quantum anomaly freedom, and interacting unitarity remain separate. A direct same-bundle Bach factorization is optional. Extending the free classification to interactions, Paper V formulates a deformation problem for the positive metric and the Krein involution. The positive Pais–Uhlenbeck metric deforms perturbatively at generic frequencies, but exact energy-conserving conversion processes generate obstruction classes. In the oscillator these occur on isolated rational resonance loci and can produce complex-conjugate spectral pairs; in the field theory, continuous momentum opens the corresponding branch-changing scattering shells, and the obstruction to the analytic positive-metric deformation is nonzero on a verified open shell subset. The massless perfect-square theory nevertheless possesses an exact nonlinear two-sector involution, showing that failure of an analytic positive pointed-sector deformation does not imply absence of a doubled Krein-type structure. Paper VI then applies the same deformation complex to the covariant Einstein–Weyl real forms of Paper IV. Cubic order is protected by the exact Einstein subsector and equal-mass kinematics, but the open process $M + M \rightarrow M + h$ produces a nonzero second-order shell cocycle. The canonical particle-number-diagonal Krein lift does not make this block null, and the verified gravitational vertices forbid a uniform abelian charge analogue of the scalar $O(1,1)$ mechanism. These results exclude neither indefinite Krein evolution nor nonanalytic, nonlocal, differently pointed, boundary-restricted, or nonperturbative completions. They show instead that the free ghost problem and the interacting probability problem are distinct, and that the latter is controlled by physical on-shell conversion processes.

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1 Why higher derivatives create ghosts

Most familiar equations of motion contain at most two time derivatives. Newton’s equation has the form

$$m\ddot{q} = F(q), \tag{1}$$

and the Klein–Gordon field equation has the form

$$(\square + m^2)\phi = 0, \tag{2}$$

where the empty square is the standard symbol for the wave operator (d’Alembertian), $\square = \partial^2/\partial t^2 - \nabla^2$ — the spacetime analogue of the Laplacian. Specifying a solution requires two pieces of initial data: position and velocity for a particle, or the field and its first time derivative for a field theory.

A fourth-order equation such as

$$\left(\frac{d^2}{dt^2} + \omega_1^2\right)\left(\frac{d^2}{dt^2} + \omega_2^2\right)z(t) = 0 \tag{3}$$

requires four pieces of initial data. Its solution contains two independent frequencies,

$$z(t) = A_1 e^{-i\omega_1 t} + B_1 e^{i\omega_1 t} + A_2 e^{-i\omega_2 t} + B_2 e^{i\omega_2 t}. \tag{4}$$

The extra derivatives have introduced an extra oscillator.

That sounds harmless. The difficulty is that the standard Hamiltonian formulation usually assigns opposite signs to the two oscillators:

$$H \sim H_{\omega_1} - H_{\omega_2}. \tag{5}$$

One branch then appears to carry negative energy. If one changes the quantization convention to keep the energy positive, the same branch may instead acquire negative norm. This is the traditional higher-derivative ghost problem.

The classical theorem of Ostrogradsky [1] explains why the Hamiltonian obtained from a nondegenerate higher-derivative Lagrangian is generically unbounded in its canonical momenta (see [2] for a modern review). But that theorem does not by itself answer every quantum question. The questions divide into three logically distinct levels.

Level 1 — local free dynamics and completion. Can the Hamiltonian be diagonalized? Can a positive inner product be found? Which variables count as real? What happens at the Jordan limit where distinct frequencies or masses coincide?

Level 2 — gauge and residual reduction. After the local gauge constraints have been imposed, which oscillator modes survive as genuine states or observables? On backgrounds with residual symmetries, should those symmetries be retained as global charges or treated as gauge? If they are gauged, what is the resulting BRST cohomology, and does positivity belong to propagating states, to vertex classes, or to neither?

Level 3 — interaction stability. Can a free positive metric or Krein involution be deformed when interactions are added? Can ghost parity remain conserved when particles convert into one another? Do exact energy-conserving processes obstruct the deformation? Does a doubled symmetry survive even if an individual pointed sector fails?

These levels must not be conflated. Solving the free Hamiltonian settles neither the residual gauge problem nor the interacting ghost problem. Residual reduction can remove locally propagating oscillator classes without changing their local equations, while interactions test whether a chosen free completion survives energy exchange. The research programme therefore has two branches after the free gravity classification: Papers V–VI study interaction stability, whereas Paper VII studies residual conformal cohomology at the pure-Weyl boundary.

Series map. Paper I reconstructs and classifies the known positive free PU quantization. Paper II proves that its canonical metric is the least-distorting free representative. Paper III separates metric, vacuum, representation, and Jordan data in the scalar field and identifies the common free complex covariance. Paper IV adds local gauge reduction and Lorentz covariance in free Einstein–Weyl gravity and follows the theory to its pure-Weyl Jordan boundary.

Two complementary branches then emerge. Papers V–VI ask whether the free real-form completions survive interaction. Paper V formulates the metric and involution deformation problem and identifies resonant on-shell obstructions; Paper VI combines that deformation complex with the covariant gravity classification of Paper IV and applies it to the complete tree-level cubic and quartic Einstein–Weyl vertex content.

Paper VII instead returns to the free pure-Weyl boundary and changes the question from completion of the local oscillator dynamics to residual gauge cohomology. On the conformal cylinder, and under the explicit choice to treat all fifteen residual conformal transformations, including compact time translation, as gauge, it derives the minimal algebraic BV complex from the pure-Weyl master action, contracts the trace, gauge-fixing, and nonminimal sectors, transfers the zero modes to one residual BFV cotangent system, and constructs the positive-frequency state complex. Its vacuum and one-particle centered sectors are acyclic, while the lowest two-particle sector contains exactly two classes with field-induced pairing

$$H_{\text{res,min}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2. \quad (6)$$

These are ghost-dressed vertex or deformation classes rather than one-particle graviton states. Descent identifies their parity combinations with the Weyl-square and Pontryagin directions; quotienting locally variationally trivial bulk functionals leaves one positive representative dynamical direction.

The residual cylinder BRST framework, centered four-ghost polarization, and weight-four selection build on Hamada and Horata’s conformal-cylinder construction [8, 9]. Paper VII’s new contribution is the completed classical chain from metric BV fields through the residual BFV

polarization to the parity-complete absolute kernel-and-image calculation: the one-particle sector vanishes, exactly two two-particle classes survive, and their field-induced pairing and dynamical/topological split are fixed.

Thus Papers I–IV classify and geometrically organize the local free theory, Papers V–VI test its interacting continuation, and Paper VII determines what survives a further residual conformal reduction at the pure-Weyl boundary. Here “canonical” and “optimal” in Papers I–II are kinematic free-theory statements, and “positive” in Paper VII refers to the normalized pairing on residual vertex representatives, not to a positive graviton Fock space.

Fourth-order theories can admit several inequivalent real-form completions of one complex spectral dynamics. Local free geometry determines which completions are available; residual gauge reduction determines which state and vertex classes survive; physical interactions determine whether the surviving structures remain dynamically compatible.

Strength of the claims. The free classifications are theorems in the classes stated below. In Paper V the 3:1 and 3:2 oscillator obstruction classes are exact, the 5:1 fourth-order result is a computational proposition, and the general resonance hierarchy remains a conjecture. For the field theory what is proved is an analytic metric obstruction on a nonempty open shell subset, not a complex spectrum or the absolute nonexistence of every positive completion. In gravity, Paper VI proves first-order cubic protection and a tree-level second-order obstruction based at the canonical Paper-IV positive real form; it does not prove an all-orders no-go, complex gravitational spectrum, or failure of every nonlocal or differently pointed completion. Its Krein result concerns the natural particle-number-diagonal, cluster-multiplicative lift induced on physical BRST cohomology, not every Poincaré-covariant involution on a multiparticle space. A full-field complex spectrum and a quantum operator implementing the doubled scalar exchange remain open.

Paper VII proves an exact residual $SO(4, 2)$ cohomology theorem in the specified quadratic algebraic cylinder category, with the chain now derived from the metric field variables onward. The smooth metric-to-curvature bridge follows from the flat BGG fine resolution and the topology of $\mathbb{R} \times S^3$; explicit all-energy $E/A/L$ preimages give the algebraic cylinder realization; the minimal master-action chain, trace contraction, gauge-fixed extension, endpoint duality, $\lambda = +1$ BV-to-BFV suspension, polarized row ledger, strict residual CE transfer, and field pairing are all explicit. The physical choice to constrain all fifteen generators, especially D , remains part of the theorem’s hypotheses. The natural energy-mode one-particle and bosonic Fock Krein completions, closed residual BRST operator, bounded Cartan contraction, and unchanged completed H^4 are now proved. The theorem does not establish an integrated $SO(4, 2)$ group representation, a positive graviton Fock space, nonlinear stability, quantum BRST nilpotency, anomaly cancellation, or interacting unitarity. The reduced Lorentzian tensor and vector fields do now carry a field-induced Cauchy–Sobolev topology Krein-unitarily equivalent to the mode completion; an ordinary-derivative realization now supplies the exact four-row symbol witness and 66-to-30 Fourier-complex retract. The curved lower-order witness/retract and adjoint table, its causal homotopies, the complete covariant/Cauchy pairing comparison, and distributional theory remain open. Direct same-bundle factorization is optional. Common-core commutators of unbounded generators do not by themselves supply the missing group integrability conditions [19].

Component	Status	Category
Smooth BGG/deformation-complex bridge	Proved	Smooth complexified cylinder fields
Metric-to-Weyl curvature isomorphism	Proved	Smooth global quotient
Explicit $E/A/L$ cylinder realization	Proved	D -finite, $SO(4)$ -finite
Minimal pure-Weyl BV/raw-chain identification	Proved	Quadratic algebraic cylinder
Trace-sector contraction	Proved	Quadratic algebraic cylinder
Gauge-fixed/nonminimal contraction	Proved	Specified algebraic gauge fixing
Fifteen zero-mode preservation	Proved	Minimal and gauge-fixed complexes
Endpoint duality	Proved	Finite algebraic blocks
Endpoint-to-Taub obstruction map	Proved	Quadratic obstruction theory
BV-to-BFV suspension	Proved, $\lambda = +1$	Selected closed-universe problem
Positive-frequency state ledger	Proved	Algebraic polarization
Strict residual CE transfer	Proved	Centered coefficient window
Two centered classes	Proved	Classical polarized state complex
Field-induced Gram matrix I_2	Proved	Selected pairing conventions
Energy-mode one-particle/Fock Krein completion	Proved	Infinite-index cylinder-energy completion
Closed residual BRST and completed H^4	Proved	Maximal total-degree block realization
Reduced Lorentzian tensor/vector Green factorization	Proved	Physical cylinder field equations
Field-induced branch Cauchy-Sobolev realization	Proved	$H^1 \oplus L^2$ on E, L ; $H^{3/2} \oplus H^{1/2}$ on A
Auxiliary symbol witness and 66-to-30 Fourier SDR	Proved	Exact cylinder symbols; finite differential and pointwise support-local formulas
Curved auxiliary witness/covariant pairing/Hadamard completion	Open	Analytic
Nonlinear stability	Open	Classical interacting theory
Quantum master equation/anomalies	Open	Quantum
Quantum survival of the two classes	Open	Quantum residual theory
Particle or S -matrix unitarity	Not claimed	Quantum interpretation

Table 1: Theorem boundary of the pure-Weyl residual branch. “Proved” in the algebraic rows always means the stated D -finite, $SO(4)$ -finite category and the selected closed-universe BFV polarization.

2 The Pais–Uhlenbeck oscillator as a microscope

The Pais–Uhlenbeck (PU) oscillator [3] is the simplest model containing the full fourth-order difficulty. Its Lagrangian can be written

$$L = \frac{\gamma}{2} (\ddot{z}^2 - (\omega_1^2 + \omega_2^2) \dot{z}^2 + \omega_1^2 \omega_2^2 z^2), \quad \omega_1 > \omega_2 > 0. \quad (7)$$

Its equation of motion factorizes:

$$\left(\frac{d^2}{dt^2} + \omega_1^2 \right) \left(\frac{d^2}{dt^2} + \omega_2^2 \right) z = 0. \quad (8)$$

It therefore contains two ordinary frequencies, but their canonical combination has the wrong-sign structure characteristic of a ghost.

The oscillator is useful because it is finite dimensional and exactly solvable. Questions that become difficult analytic problems in a field theory reduce here to matrix algebra:

- Can the Hamiltonian be transformed to a positive normal form?
- Is that transformation unique?
- How many positive metrics exist?
- Is there a preferred one?
- What happens as $\omega_1 \rightarrow \omega_2$?
- Does the limiting Jordan system admit any positive invariant inner product?

A correct answer at the oscillator level does not automatically solve a quantum field theory. But the oscillator provides the local building block for each momentum mode of a free fourth-order field.

3 What Bender and Mannheim had already solved

Bender and Mannheim [5, 6] observed that the PU Hamiltonian should not be quantized using the naive real contour and inner product. After a complex continuation of one canonical coordinate, the Hamiltonian becomes non-Hermitian but PT-symmetric [4].

They constructed an operator

$$Q = \alpha pq + \beta xy \quad (9)$$

and the associated similarity transformation $\rho = e^{-Q/2}$ such that

$$h = \rho H_{\text{PT}} \rho^{-1} \quad (10)$$

is a conventional positive Hamiltonian describing two ordinary oscillators. The original Hamiltonian is then Hermitian not with respect to the naive adjoint, but with respect to the positive metric

$$\eta = \rho^\dagger \rho = e^{-Q}, \quad (11)$$

in the general framework of pseudo-Hermitian quantum mechanics [20, 21]. The physical inner product becomes

$$\langle \psi, \phi \rangle_\eta = \langle \psi, \eta \phi \rangle. \quad (12)$$

With this inner product, the unequal-frequency oscillator has positive norms and a positive spectrum. Thus an important version of the ghost problem was already solved:

The free unequal-frequency Pais–Uhlenbeck oscillator admits a positive-metric quantization.

Three levels must nevertheless be distinguished, and the series addresses each in turn: the *free spectral solution* (the free oscillator maps to a positive two-oscillator Hamiltonian); *perturbative interaction stability* (the metric can be deformed away from on-shell resonances); and *global interacting viability* (energy-conserving conversion processes may obstruct the deformation and, in finite resonant systems, produce complex spectra). In short:

The free ghost problem was solved locally; Paper V determines where that solution survives interaction.

What remained unclear was the mathematical status of the construction. Why does the particular operator Q appear? Is it an ansatz or a canonical object? Is the metric unique? If it is not unique, why should the Bender–Mannheim metric be preferred? What happens when the oscillator becomes a field with infinitely many momentum modes? How does this positive construction relate to an indefinite Krein-space construction [7]? And what survives after gauge reduction in gravity? The four papers [25, 26, 27, 28] address these questions.

4 Paper I: reconstructing the Bender–Mannheim transformation

The first paper [25] reformulates the oscillator as a problem in complex symplectic geometry. Collect the canonical variables into a vector $V = (x, y, p, q)^\top$, with commutation relations encoded by the symplectic matrix J :

$$[V_a, V_b] = iJ_{ab}. \quad (13)$$

A quadratic Hamiltonian has the form $H = \frac{1}{2}V^\top GV$, where G is a symmetric matrix, and the Hamiltonian flow is generated by $A = JG$. The problem is to find a complex symplectic matrix S satisfying

$$S^\top JS = J, \quad S^\top GS = G_0, \quad (14)$$

where G_0 is the positive two-oscillator normal form.

The first result is that, after a specified canonical normalization, this problem has a *unique* Hermitian positive-definite solution S_+ . The complete solution family is the coset

$$S_+H, \quad H \cong SO(2, \mathbb{C}) \times SO(2, \mathbb{C}), \quad (15)$$

where H is the stabilizer of the positive normal form. Thus many complex symplectic transformations diagonalize the Hamiltonian, but exactly one normalized representative is both Hermitian and positive.

The logarithm of this matrix is then converted into a quadratic quantum operator through the complexified metaplectic representation [32]. The resulting operator is precisely $Q = \alpha pq + \beta xy$ with the coefficients found by Bender and Mannheim. The logical direction has therefore been reversed:

$$(G, J, G_0) \longrightarrow S_+ \longrightarrow Q.$$

The operator Q is no longer introduced by guessing a useful quadratic expression. It is reconstructed from the symplectic normal-form problem.

4.1 The exact spectrum of Q

In normalized oscillator coordinates,

$$Q = r (a_1 a_2^\dagger + a_1^\dagger a_2), \quad r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}. \quad (16)$$

This is a number-preserving Schwinger $SU(2)$ generator. It commutes with $N_{\text{tot}} = N_1 + N_2$, and on the fixed- N subspace its eigenvalues are $r(-N, -N+2, \dots, N-2, N)$. Consequently

$$\text{spec } Q = r \mathbb{Z}, \quad (17)$$

with pure-point spectrum and infinite multiplicities. The metric has spectrum

$$\text{spec}(e^{-Q}) = \{e^{-rn} : n \in \mathbb{Z}\} \cup \{0\}, \quad (18)$$

where zero is an accumulation point rather than an eigenvalue. This provides a natural invariant domain: the algebraic Fock space of finite linear combinations of number states, preserved by Q , $e^{\pm Q/2}$, and the quadratic Hamiltonians appearing in the construction.

4.2 Many metrics, one canonical Gaussian metric

Pseudo-Hermiticity alone does not make the metric unique. If $h = \rho H_{\text{PT}} \rho^{-1}$ and W is a positive operator commuting with h , then

$$\eta_W = \rho^\dagger W \rho \quad (19)$$

also defines an algebraic positive metric form [20, 21]. Thus the complete metric freedom is larger than the finite-dimensional symplectic stabilizer: the stabilizer describes the Gaussian or quadratic part of the ambiguity, while arbitrary positive functions of the normal-mode number operators provide additional non-Gaussian freedom. The first paper therefore distinguishes the *unique normalized positive symplectic diagonalizer* from the *nonunique pseudo-Hermitian metric operator*.

5 Paper II: why the canonical metric is the least-distorting one

Uniqueness inside a restricted class does not by itself explain why S_+ is geometrically natural. The second paper [26] supplies a variational characterization. For an admissible diagonalizer S , define

$$\mathcal{F}(S) = \|\log(S^\dagger S)\|_F^2. \quad (20)$$

The positive matrix $S^\dagger S$ measures how much the transformation stretches and contracts the canonical coordinates; the logarithm converts multiplicative stretching into additive “rapidities,” and the Frobenius norm measures their total size. The theorem is

$$\mathcal{F}(S) \geq \mathcal{F}(S_+) = 4r^2, \quad (21)$$

with equality precisely for S_+ multiplied by a unitary element of the stabilizer — which for the PU oscillator consists of real sign matrices, so all minimizers have the same positive polar factor and determine the same Bender–Mannheim metric. In ordinary language:

S_+ is the least-distorting allowed transformation that makes the Hamiltonian positive.

This is analogous to finding the shortest path from a point to a surface: a shortest path meets a smooth surface orthogonally. Here the “surface” is the orbit generated by transformations that preserve the positive normal form.

5.1 Cartan projection

The geometry is that of the complex symplectic group $\mathrm{Sp}(4, \mathbb{C})$ and its maximal compact subgroup $K = \mathrm{Sp}(4, \mathbb{C}) \cap U(4)$, acting on the space of positive matrices with the affine-invariant metric [33]. The Cartan projection records the logarithmic singular values of a transformation. With the doubled Gram convention used in the papers ($\mu(S) = \log \mathrm{spec}(S^\dagger S)$),

$$\mu(S_+) = (r, r), \quad (22)$$

a point on the wall $x_1 = x_2$ of the C_2 Weyl chamber. The distortion functional is related to the Cartan norm by

$$\mathcal{F}(S) = 2\|\mu(S)\|^2. \quad (23)$$

Thus the minimum-distortion theorem is a Cartan-projection theorem.

5.2 Why the theorem is not a routine Kempf–Ness application

The physical stabilizer $H \cong \mathrm{SO}(2, \mathbb{C})^2$ is not compatible with the canonical Cartan involution determined by the physical inner product: generically

$$H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \quad (24)$$

rather than the maximal compact $U(1)^2$ of the abstract group. The obstruction belongs to the *embedding* of H , not to its abstract group structure, so the classical nearest-point theory of self-adjoint groups [34, 35] does not apply to H directly.

The resolution is to enlarge H to its smallest compatible (θ -stable) hull. Generically this is

$$SL(2, \mathbb{C}) \times SL(2, \mathbb{C}), \quad (25)$$

the mode-preserving subgroup, whose positive part is a totally geodesic copy of $\mathbb{H}^3 \times \mathbb{H}^3$. The Bender–Mannheim direction is orthogonal to this mode-preserving geometry, so standard nonpositively curved projection theory can be applied to the hull and the result restricted back to the physical stabilizer. The paper proves a general principle:

A positive diagonalizer minimizes Cartan distortion along a compatible subgroup exactly when its Hermitian generator is orthogonal to the subgroup's positive tangent directions.

For a splitting into many oscillator modes, those tangent directions are the mode-preserving deformations, and the normal directions are precisely the inter-mode couplings. The PU oscillator is the simplest nontrivial example: its preferred generator is entirely inter-mode.

6 The exceptional point: when two frequencies merge

The rapidity is $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$. As $\omega_1 \rightarrow \omega_2$, the denominator tends to zero and $r \rightarrow \infty$. The positive diagonalizer stretches one pair of directions by $e^{r/2}$ and contracts another by $e^{-r/2}$; the corresponding Cartan distance diverges.

This is not merely a poor choice of coordinates. At equal frequency, the dynamical matrix ceases to be diagonalizable and develops a Jordan block — the hallmark of an exceptional point [37, 38]. A 2×2 Jordan block has the form

$$H_J = \begin{pmatrix} \omega & 1 \\ 0 & \omega \end{pmatrix}. \quad (26)$$

Suppose there were a positive-definite Hermitian form η satisfying $H_J^\dagger \eta = \eta H_J$. Then the matrix $\eta^{1/2} H_J \eta^{-1/2}$ would be Hermitian, hence diagonalizable. But similarity transformations preserve Jordan form. This is impossible. Hence a nontrivial Jordan block admits no nondegenerate positive-definite invariant Hermitian form [25, 36]. It may still admit:

- degenerate positive-semidefinite forms;
- nondegenerate indefinite forms;
- or intrinsically Jordan PT-symmetric realizations [6].

What fails is the continuation of the unequal-frequency positive Hilbert metric as a nondegenerate invariant positive form. This distinction matters: the conclusion is not that the equal-frequency theory cannot exist, but that it cannot remain an ordinary diagonalizable positive-metric system of the same kind.

7 From one oscillator to a field

Consider the free fourth-order scalar field

$$(\square + m_1^2)(\square + m_2^2)\phi = 0. \quad (27)$$

After Fourier transforming in space, each spatial momentum k contributes a PU oscillator with frequencies $\omega_i(k) = \sqrt{k^2 + m_i^2}$. The modewise rapidity is

$$r(k) = \log \frac{(\omega_1(k) + \omega_2(k))^2}{m_1^2 - m_2^2} = 2 \log |k| + O(1) \quad (|k| \rightarrow \infty). \quad (28)$$

Thus the modewise similarity transformations become increasingly nonuniform in the ultraviolet.

This is where finite-dimensional and field-theoretic questions diverge. Each individual momentum mode can be treated successfully, while the collection of infinitely many modes may define an unbounded field transformation, a non-equivalent norm on phase space, a different quasifree representation, or a different real form of the field algebra. A modewise solution is therefore necessary but not sufficient for a field theory.

8 Metric geometry and vacuum geometry are different

A central lesson of the field papers [26, 27] is

$$\text{metric distortion} \neq \text{vacuum excitation.}$$

The Cartan rapidity $r(k)$ diverges like $2 \log k$, and the finite-dimensional observable-space metric has eigenvalues $e^{\pm r(k)} \sim k^{\pm 2}$. This describes a strong deformation of the canonical phase-space norm. It does *not* follow that the transformation creates $O(k^2)$ particles.

In normalized coordinates the generator is number preserving:

$$Q_k = r(k)(a_{1,k}a_{2,k}^\dagger + a_{1,k}^\dagger a_{2,k}). \quad (29)$$

It acts inside the complexified beam-splitter algebra, and the canonical vacuum is annihilated by it, even though the associated Cartan norm is arbitrarily large.

The reason is geometric. Positive metrics are points in one quotient space, roughly G/K , while Gaussian vacuum rays belong to a different quotient, G/P_Ω , where P_Ω stabilizes the vacuum line. The vacuum-ray map does not factor through the metric quotient. Two transformations can therefore have very different metric distortion while producing the same vacuum ray. This prevents one from estimating physical particle content directly from the Cartan norm.

9 The universal complex vacuum covariance

For the fourth-order field, the selected two-point function has the spectral form

$$\widetilde{W}_{12}^+(p) = \varepsilon \theta(p^0) \frac{\delta(p^2 - m_1^2) - \delta(p^2 - m_2^2)}{m_1^2 - m_2^2}, \quad (30)$$

where $\varepsilon = \pm 1$ records the overall action convention. This is a divided difference of two ordinary Klein–Gordon two-point functions:

$$W_{12}^+ = \varepsilon \frac{W_{m_1}^+ - W_{m_2}^+}{m_1^2 - m_2^2}. \quad (31)$$

The field paper [27] shows that the admissible positive metrics all select the same complex covariance. They may change the physical adjoint and the representation of observables, but not this underlying spectral two-point distribution. The result should be stated carefully:

The complex covariance is metric-independent; the involution and completion need not be.

A state in the strict algebraic sense requires a fixed involution and a positivity condition. When different completions use different involutions, it is safer to speak first of the common complex quasifree covariance.

10 Resonance: the metric diverges but the covariance survives

Let $m_1^2, m_2^2 \rightarrow m^2$. Then the divided difference has the regular limit

$$\widetilde{W}_{mm}^+(p) = -\varepsilon \theta(p^0) \delta'(p^2 - m^2). \quad (32)$$

In the time domain, per mode,

$$W_{mm}^+(t, k) = -\varepsilon \frac{(1 + i\omega t) e^{-i\omega t}}{4\omega^3}, \quad \omega = \sqrt{k^2 + m^2}. \quad (33)$$

The positive diagonalizing transformation diverges because the dynamics becomes Jordan. Yet the complex vacuum covariance remains finite. This is one of the main conceptual results:

Resonance is a singularity of the metric and dynamics, not necessarily of the vacuum covariance.

At the doubly massless point $m = 0$, the covariance is related, after matching the action sign and infrared convention, to the massless $\square^2$ covariance used in the Krein-space construction of Bateman and Turok [7]. Their sign convention gives the opposite overall covariance to one common PU convention: the relation is an action-sign reversal, not a literal equality before conventions are matched. This bridge identifies the *free complex covariance*; it does not by itself reconstruct Bateman and Turok's interacting $O(1,1)$ embedding, ghost symmetry, or indefinite-space probability rule, which are additional structures of their theory.

11 What is a Krein space?

A Hilbert space has a positive inner product: $\langle \psi, \psi \rangle > 0$ for every nonzero state. A Krein space has a nondegenerate but *indefinite* inner product: a state can have positive, negative, or zero norm.

At first sight this appears incompatible with probability. But indefinite-metric quantum theories can possess an additional grading or symmetry that identifies the physical probability structure. In the Bateman–Turok construction [7], a hidden ghost-parity symmetry plays this role.

The positive and Krein descriptions sacrifice different conventional requirements. For an unpaired ghost oscillator, one may try to demand:

1. positive norm;
2. positive energy;
3. standard reality of the original field.

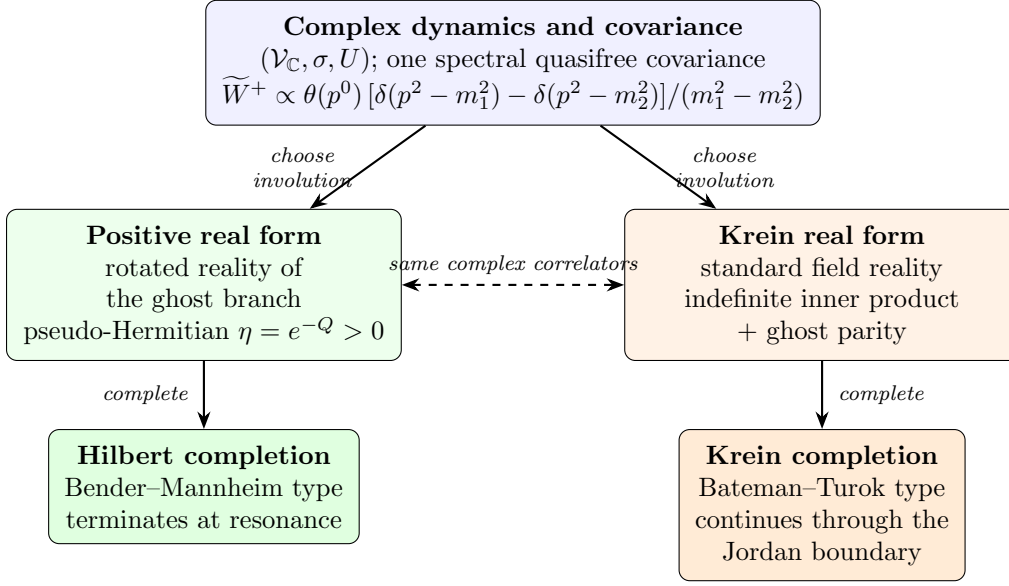


Figure 1: The three-level structure: one complex spectral covariance, two covariant real forms (involutions), two completions. The two branches agree on all complex correlation functions and differ in which operators are Hermitian and how probabilities are represented. At the resonant (Jordan) boundary only the Krein branch continues as a nondegenerate completion.

The three cannot all be retained simultaneously [28]. A complex quarter-turn of the ghost variables can make the energy and norm positive, but the original amplitude becomes anti-Hermitian; i times the amplitude is the physical Hermitian observable. Alternatively, the original field can remain real and the energy spectral condition can be retained, but the state space becomes indefinite. This gives two real forms of a common complex system:

	positive completion	Krein completion
energy	positive	positive
norm	positive	indefinite
original field reality	rotated	standard

The complex equations and complex covariance may agree even though the physical adjoints differ. Figure 1 summarizes the resulting three-level structure.

12 A necessary representation-theoretic distinction

There are two different ways to compare the positive PT construction with an ordinary oscillator Fock space, and conflating them produces contradictions [26].

12.1 Physical Dyson transport

For each mode, the PT ground state is exactly the Dyson pullback of the positive normal-form vacuum,

$$\psi_k = \rho_k^{-1} \phi_k. \quad (34)$$

Equip the PT space with the metric $\eta_k = \rho_k^\dagger \rho_k$. Then

$$\rho_k : \mathcal{H}_{\text{PT},k} \longrightarrow \mathcal{H}_{\text{normal},k} \quad (35)$$

is unitary with respect to the physical inner products, and $\rho_k \psi_k = \phi_k$ *exactly*. The pointed modewise unitaries therefore tensor together without any vacuum-overlap obstruction: the completed positive PT theory is globally equivalent to its positive normal-form theory.

12.2 Identity embedding in an auxiliary standard CCR algebra

One may instead place both Gaussian wavefunctions in the same standard Schrödinger L^2 space, retain the same canonical variables and standard adjoint, and compare the vectors through the identity embedding. Under that comparison, the per-mode Gaussian fidelity approaches $\sqrt{3}/2$, the expectation of the standard auxiliary number operator approaches $1/3$, and the infinite products of these auxiliary standard-CCR states are disjoint [39, 26].

This does not contradict the physical Dyson equivalence: the two comparisons identify observables differently.

identity embedding of auxiliary variables $\neq$ Dyson transport of the physical algebra.

This is an example of why field representation theory cannot be discussed without specifying the algebra, its involution, the embedding between descriptions, and the chosen completion.

13 The ultraviolet Fock-sector hierarchy

The preceding comparison placed a selected PU vacuum beside the split two-field vacuum. Paper III also asks a different question: when do two *selected* vacua, belonging to different mass pairs (m_1, m_2) and $(\bar{m}_1, \bar{m}_2)$, define the same ultraviolet Fock sector in common field variables? Writing $\Sigma = m_1^2 + m_2^2$ and $\Pi = m_1^2 m_2^2$, the modewise fidelity obeys

$$1 - f(k) = \frac{(\Sigma - \bar{\Sigma})^2}{12k^4} + O(k^{-6}). \quad (36)$$

If $\Sigma = \bar{\Sigma}$, the k^{-6} term also vanishes and the first remaining obstruction is proportional to $29(\Pi - \bar{\Pi})^2/(576k^8)$. Since the two symmetric invariants (Σ, Π) determine the unordered mass pair, the ultraviolet classification terminates:

spatial dimension	ultraviolet sector invariant
$d \leq 3$	none
$4 \leq d < 8$	$\Sigma = m_1^2 + m_2^2$
$d \geq 8$	(Σ, Π) , the unordered mass pair

At $d = 4$ and $d = 8$ the corresponding obstruction is logarithmic, so the endpoints belong to the higher-dimensional rows shown.

This is an *ultraviolet* classification, not automatically a global GNS-equivalence theorem. For strictly positive masses the infrared is regular. Against the doubly massless $\square^2$ vacuum, however, the relative occupation behaves as k^{-3} and its infrared integral diverges for $d \leq 3$. Whether the massless anchor belongs to the same global sector in those dimensions therefore remains open. This is why “one ultraviolet sector” must not be strengthened to “one global representation.”

14 The fourth-order vacuum is locally well behaved

Ordinary four-dimensional Klein–Gordon two-point functions have a leading short-distance singularity proportional to $1/\sigma_+$, where $\sigma_+(x, y)$ is the regularized squared geodesic separation. In the fourth-order divided difference, this mass-independent leading term cancels. The result has the Hadamard-like form

$$W_{12}^+(x, y) = V_{12}(x, y) \log \sigma_+(x, y) + H_{12}(x, y), \quad (37)$$

where V_{12} and H_{12} are smooth and

$$V_{12}(x, x) = \frac{\varepsilon}{16\pi^2}. \quad (38)$$

There are further terms such as $\sigma \log \sigma$, so the two-point function is not simply a constant logarithm plus a smooth remainder. Nevertheless, its wavefront directions are the standard positive-frequency Hadamard directions, in the microlocal sense of [40].

This is the natural fourth-order analogue of the Hadamard property [27]. It says that the global ghost or completion problem does not automatically produce a pathological local ultraviolet singularity: the vacuum can be globally unusual in its real form or representation, yet locally well behaved in the microlocal sense required for renormalized observables [41, 42].

15 When interactions are switched on

Everything so far concerns free theories. The fifth paper of the series [29] asks the next question: can the free positive or Krein structure be continuously deformed when nonlinear processes are switched on? The answer has four parts: generically yes at low perturbative order away from resonances; no on specific energy-conserving conversion channels; in field theory continuous momentum makes such channels kinematically available for generic split masses, with the obstruction verified on a nonempty open shell subset; and an exact doubled sector-exchange symmetry can survive even though the analytic deformation of the canonical positive pointed-sector metric fails there. Figure 2 summarizes the mechanism.

15.1 Deforming the positive metric

Perturb the free theory, $H_\zeta = H_0 + \zeta V$, and recall the free Dyson map with $\rho_0 H_0 \rho_0^{-1} = h_0 = h_0^\dagger$. One seeks a corrected map

$$\rho_\zeta = e^{-\frac{1}{2}(\zeta R_1 + \zeta^2 R_2 + \dots)} \rho_0, \quad (39)$$

keeping the transformed Hamiltonian Hermitian order by order. At first order,

$$[h_0, R_1] = v^\dagger - v, \quad v = \rho_0 V \rho_0^{-1}. \quad (40)$$

The interaction introduces a new “bad” non-Hermitian part, and the correction R_1 asks whether the measuring geometry can be adjusted to absorb it. The key principle is simple: a correction can remove terms that move states between *different* energies; it cannot remove a term that acts entirely among states with exactly the *same* energy. When two states have different energies, the metric can distinguish them; when they have exactly equal energy,

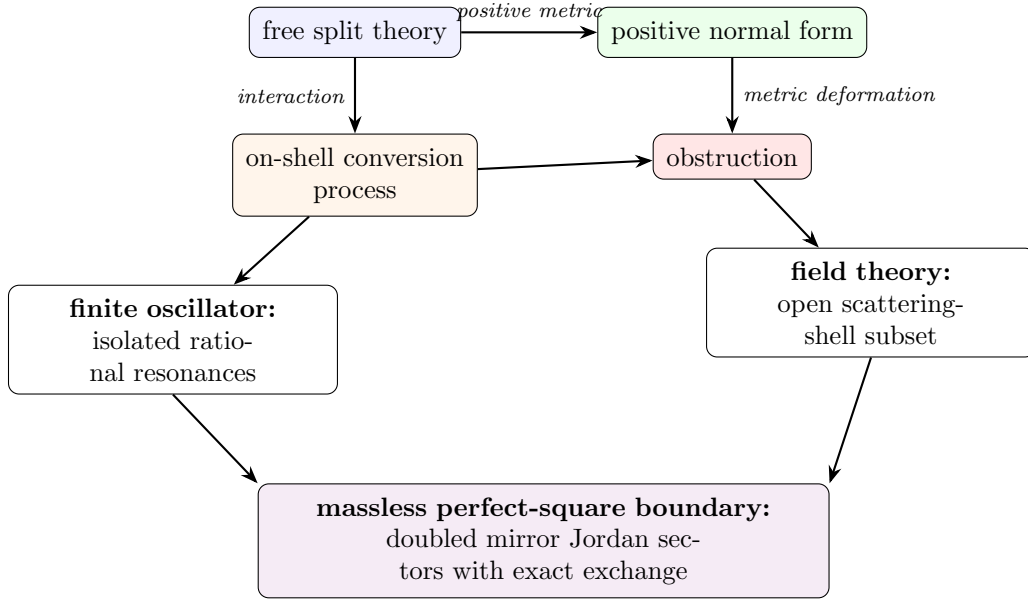


Figure 2: The interaction mechanism of [29]: metric deformation meets on-shell conversion; isolated resonances in the oscillator become kinematically available scattering shells in the field theory, with the obstruction verified on a nonempty open subset; at the massless boundary a doubled exchange structure survives.

the interaction is resonant and cannot be rotated away. Technically, the obstruction is the energy-conserving part of the source,

$$\mathfrak{o}_+ = \Pi_{\ker \text{ad}_{h_0}} S, \quad (41)$$

the projection onto processes satisfying exact free-energy conservation.

15.2 A finite oscillator fails only at special resonances

For the cubic PU interaction at generic unequal frequencies, the positive metric can be corrected through the computed orders — the explicit generators exist and the transformed Hamiltonian is Hermitian. But at

$$\omega_1 = 3\omega_2, \quad (42)$$

one high-frequency quantum can convert into three low-frequency quanta without changing the total energy, and the second-order correction is obstructed by exactly that conversion operator, $a_1 a_2^{\dagger 3} - a_1^{\dagger} a_2^3$. At $2\omega_1 = 3\omega_2$ a third-order obstruction appears. A transfer-lattice rule organizes which conversion monomials can first occur at each order. Its first new prediction has been confirmed computationally: at $\omega_1 = 5\omega_2$ the second- and third-order classes vanish, while the fourth-order class is a nonzero $1 \leftrightarrow 5$ conversion proportional to $a_1 a_2^{\dagger 5} - a_1^{\dagger} a_2^5$. This confirms one prediction of the proposed hierarchy; the general $p:q$ rule remains a conjecture.

Imagine two musical notes: usually they are unrelated, but if one note has exactly three times the frequency of the other, one vibration can split into three without changing the total energy — at that point the interaction is perfectly synchronized with the system.

The 3:1 obstruction is more than a failure of one formula. Sufficiently excited resonant multiplets develop complex-conjugate energies, $E_{\pm} = E_R \pm i\Gamma$, at second order. This is formally defined perturbative PT breaking: [29] constructs the antilinear symmetry explicitly, proves that it commutes with the interacting Hamiltonian, and certifies the complex pair *exactly* in the effective resonant multiplet (by rational algebra and Sturm counting, not only numerically). Where the complex pair persists, no positive-definite invariant metric can make the Hamiltonian equivalent to an ordinary self-adjoint one. Two levels of conclusion must not be merged: this complex-spectrum statement is established in the verified resonant oscillator setting; the field-theory result below is presently a perturbative metric obstruction, not yet a demonstrated complex spectrum.

15.3 The momentum continuum turns exceptional resonances into ordinary scattering

For an oscillator, the frequencies are fixed numbers and exact resonances require special ratios. For a field, $E_a(k) = \sqrt{k^2 + m_a^2}$, and momenta vary continuously: one can satisfy

$$E_H(k_1) + E_L(k_2) = E_L(k_3) + E_L(k_4) \quad (43)$$

together with momentum conservation on open sets of scattering data, for *any* split masses. The key process is the branch-changing scattering $H + L \rightarrow L + L$. Paper V shows that the second-order positive-metric source is nonzero on an open scattering-shell subset (exactly, at a rational kinematic point, hence on an open neighborhood — not claimed at every kinematic point): the continuum turns the oscillator’s isolated resonance obstruction into an obstruction on a nonempty open part of a scattering shell. In a small mechanical system, resonance requires the machine to be tuned to a special ratio; in a field, particles can change their momentum, and even if their masses are not specially tuned, their motion can supply the exact energy match.

The canonical positive completion therefore remains meaningful locally and perturbatively away from resonant processes, but it cannot be deformed analytically into a positive pointed-sector metric on the verified branch-changing shell subset. One qualification is essential: this establishes that *no analytic deformation of the canonical metric* exists there — not that no conceivable positive completion of any kind exists. For the oscillator with certified complex energies the stronger conclusion is available; for the field theory the perturbative statement is what is proved.

15.4 Why the obstruction waits until second order

The perfect-square cubic vertex is exactly the Källén polynomial, $\mathcal{V}_3 = \frac{1}{2}\lambda_K(p_1^2, p_2^2, p_3^2)$ (Section 14 notation). It vanishes for three ordinary massless legs and has a second-order zero at the confluent massless point; and the transported reality factors eliminate the first-order positive obstruction on the physically open decay channel. So the positive metric survives the first test. But at second order, two cubic interactions combine into a branch-changing $2 \rightarrow 2$ process, and the obstruction appears: a single interaction event is too constrained to cause the problem, while two interaction events in sequence can convert the particles while preserving total energy — and that is where the positive geometry fails.

15.5 A different structure survives: two mirror Jordan sectors

The perfect-square theory admits an exact rewriting, introduced by Bateman and Turok [7], who also identified its exchange symmetry as ghost parity and as charge conjugation for an internal $O(1,1)$ symmetry; the sector-level reading given here is what the interaction paper adds. Introducing an auxiliary field and the nonlinear change of variables $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$,

$$\mathcal{L} = -\partial_\mu U \partial^\mu V + \frac{\lambda^2}{2} U^2 V^2. \quad (44)$$

The constant zero-potential stationary backgrounds form two branches, $(U_0, 0)$ and $(0, V_0)$, and the exact exchange $U \leftrightarrow V$ maps one branch to the other. Around $(1, 0)$ the linearized equations are $\square v = 0$, $\square u = \lambda^2 v$ — a Jordan chain generated by the interaction itself; around $(0, 1)$ the chain has the opposite orientation. The complete theory thus has two mirror valleys: choosing one valley gives one orientation of the Jordan system, and the exact symmetry does not rearrange particles inside that valley — it carries the entire valley to its mirror.

Three distinctions matter here. The exact symmetry belongs globally to the *extended* (U, V) theory; its expression in the original (ϕ, ψ) variables is a chart-transition map, singular at the chosen perturbative vacuum; and it is not a ghost-number sign inside one Fock sector. It may ultimately form part of an antiunitary CPT-like structure, but what is proved is a linear sector exchange, not a quantum CPT theorem.

Finally, the ordinary split-theory ghost parity becomes singular at the merge. In the confluent Jordan basis $c = (b_+ + b_-)/2$, $d = (b_+ - b_-)/\delta$, the branch parity becomes

$$P_\delta = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}, \quad P_\delta^2 = 1, \quad (45)$$

with no bounded limit on a single Jordan chain: as the two particle types merge, the old rule “one is even and one is odd” becomes infinitely badly conditioned. The only finite remnant is to exchange the entire Jordan system with its mirror copy: after doubling by the oppositely oriented sector, the singular parity becomes a finite sector exchange. This confluent result is the *linear bridge* to the independently exact nonlinear $U \leftrightarrow V$ symmetry: what the regulated parity converges to is the linearization of that exchange around the two mirror backgrounds — no stronger identification is claimed.

The relation between the two structures is sharp in the verified perturbative setting. At the rational shell point used to certify the field obstruction, the complete reachable second-order on-shell matrix is *exactly* pseudo-Hermitian with respect to ghost-branch parity, and its entire positive-metric obstruction lies in the parity-odd block. On the same matrix elements the Krein pairing is preserved while the analytic deformation of the canonical pointed positive metric is obstructed. The two escapes from the ghost problem therefore *separate*: the Krein completion is not a hidden version of that positive completion. This does not exclude a nonanalytic or differently pointed positive construction, nor does it prove a complex field-theory spectrum.

There is a further boundary distinction. Exact pseudo-Hermiticity of the split on-shell matrix is weaker than the one-sided charge condition used in the Bateman–Turok generalized Born rule. A canonical map to their cross-paired charge basis sends the regulated pointed vacuum to a Gaussian with

$$\frac{S_{UU}}{S_{VV}} = \left(\frac{\delta}{2g} \right)^2 = \frac{\epsilon}{g}, \quad \delta^2 = 4\epsilon g. \quad (46)$$

Thus both charge signs are intrinsic at split mass. The image becomes one-sided only on the $O(1,1)$ -symmetric confluent line $\epsilon = 0$; the Bateman–Turok massless point additionally has $\mu^2 = 0$. A self-contained charge-null lemma then makes every strictly negative component null and orthogonal to the neutral component. The remaining capstone calculation is to transport the explicit obstruction through this boundary limit and evaluate its generalized Born trace.

Terminology.

Positive pseudo-Hermitian metric: a positive inner product obtained by changing the adjoint through a Dyson map. *Metric deformation*: an order-by-order correction of that inner product after an interaction is introduced. *Obstruction*: the energy-conserving part of the unwanted non-Hermitian interaction that cannot be generated by a metric correction. *Pointed sector*: a quantum construction built around one selected vacuum background. *Doubled theory*: a completion containing both mirror Jordan sectors. *Sector-exchange involution*: the exact map exchanging the two mirror backgrounds and reversing the Jordan-chain orientation. *Spectral breaking*: the appearance of complex-conjugate energy levels, which excludes a positive invariant metric where the complex spectrum persists. “Ghost parity” is used only with a qualifier: split branch-number parity, the exact $U \leftrightarrow V$ exchange, or a (future) antiunitary CPT-like operation are different objects.

16 Why gravity is not merely five copies of the oscillator

The scalar PU oscillator has no gauge symmetry. Gravity does.

Consider linearized scalar-free Einstein–Weyl gravity around flat spacetime: the Einstein term plus a tuned quadratic-curvature term ($\alpha R_{\mu\nu}^2 + \beta R^2$ with $\alpha = -3\beta$). The linearized spectrum contains a massless graviton with two helicities and a massive spin-2 branch with five helicities [24]. A naive polarization-by-polarization treatment might suggest that every massive helicity belongs to an independent PU pair. Gauge reduction shows otherwise [28]: the reduced phase space has the structure

$$\Gamma_{\text{phys}} \cong 2\Gamma_{\text{PU}} \oplus 2\Gamma_- \oplus \Gamma_-. \quad (47)$$

The two helicity-2 sectors are genuine fourth-order PU pairs: each contains a massless graviton mode and a massive ghost partner. The helicity-1 and helicity-0 sectors contain only *unpaired* second-order massive ghosts: their apparent massless partners lie entirely in the diffeomorphism gauge orbit and disappear under reduction.

Gauge reduction preserves PU pairing only in helicity ± 2 .

This is the first major way in which gravity differs from the scalar model.

17 Lorentz covariance forbids hybrid solutions

The five massive helicities $+2, +1, 0, -1, -2$ form one irreducible massive spin-2 representation. They are not five independent species that can be quantized separately: a Lorentz transformation can mix them.

By Schur’s lemma, an invariant Hermitian form on the irreducible spin-2 polarization space is proportional to the identity. Its sign or reality assignment must therefore be uniform across

all five helicities. This rules out a tempting hybrid construction in which the helicity-2 modes receive the positive PU metric while the helicity-1 and helicity-0 ghosts are treated in a Krein space: such an assignment is not Lorentz covariant.

Within the class analyzed in the gravity paper [28] — free, translation-invariant, quasifree, mode-local constructions compatible with the reduced symplectic form, the spectral condition, and Poincaré covariance — there are exactly two covariant completion classes:

1. a positive pseudo-Hermitian completion in which the full massive spin-2 multiplet has uniformly rotated reality;
2. a Krein completion in which the gravitational field retains standard reality and the entire massive spin-2 multiplet carries a uniform negative sign relative to the positive massless graviton sector.

The two completions induce the same complex gauge-invariant covariance, while differing in their involution and completion. This is the gravitational version of the real-form distinction of Figure 1.

18 The gauge-invariant gravitational bridge

The equality of the two complex covariances is not merely a polarization-by-polarization analogy. The reduced helicity kernels have residues fixed by the reduced symplectic form. Their ratios $1 : M^2 : \frac{1}{3}$ match the gauge-invariant contractions of the massive spin-2 projector,

$$\frac{1}{2} : \frac{M^2}{2} : \frac{1}{6} \tag{48}$$

so they reassemble, modulo pure-gauge bi-distributions, with one overall coefficient $4/c_1$. The massless shell couples only to the transverse-traceless polarizations. Thus both real forms determine the same complex spectral functional on the algebra of linear gauge-invariant observables, rather than merely agreeing in a chosen helicity frame.

The massive projector itself contains apparently singular terms $p_\mu p_\nu / M^2$. Applying the linearized Weyl-tensor map annihilates all of them: the Weyl–Weyl kernel is gauge independent, covariant, and regular as $M \rightarrow 0$, with the standard positive-frequency Hadamard wavefront directions. This cancellation is why the conformal limit below is formulated on the curvature-generated regular subalgebra. The raw scalar potential kernel does not have a regular limit; the divergent direction becomes Weyl-null and must first be quotiented out.

Relation to CPT language. Mannheim’s CPT-based positive scalar product belongs, away from the conformal locus, to the positive class just described, with its uniformly rotated massive reality [23]. At the Jordan boundary the no-go theorem excludes a time-independent, nondegenerate, positive-definite invariant form; it does not exclude the intrinsically Jordan realization of Bender and Mannheim, which uses nonstationary Jordan states and a degenerate pairing [6]. The two statements are therefore compatible: the theorem delimits which positive completion terminates, rather than declaring the equal-frequency theory nonexistent.

19 The conformal Jordan boundary

Let the coefficient of the Einstein term tend to zero while the quadratic-curvature coupling remains fixed. The massive spin-2 mass then tends to zero, and the different helicity sectors behave differently [28].

Helicity ± 2 . The massless and massive branches merge, producing two $\square^2$ Jordan systems. Each polarization contributes two configuration-space solutions, giving four modes in total.

Helicity ± 1 . The vector ghosts have an ordinary massless limit. They remain second-order modes rather than becoming Jordan partners.

Helicity 0. The scalar kinetic normalization tends to zero. At the conformal point a new Weyl gauge symmetry appears, $\delta h_{\mu\nu} = 2\sigma\eta_{\mu\nu}$: the scalar direction becomes null in the presymplectic form and is removed by the quotient.

The conformal count is therefore $4 + 2 + 0 = 6$. The positive split-frequency metric runs to infinite Cartan distance and has no nondegenerate positive continuation through the Jordan boundary; the complex spectral covariance and the standard-reality Krein form can continue, provided the newly null Weyl direction is quotiented out. The algebra of observables changes rank at the boundary, so convergence must be formulated on a fixed regular family of observables — most naturally through gauge- and Weyl-invariant curvature quantities. (Boundary constructions that instead *remove* one branch of the solution space, as in [43], and the richer phase diagram at nonzero cosmological constant [44, 45], are related but distinct stories.)

20 Paper VII: what remains after residual gauge reduction?

The ghost problem is usually introduced by listing the modes that solve a higher-derivative field equation and then examining the signs of their energies or inner products. Papers I–IV analyse precisely this local free problem: which modes are present, how they transform, which real forms and metrics are available, and what happens when distinct branches merge.

In gravity, however, there is a logically prior question:

Does every mode appearing in the linearized equations represent an independent physical object, or is some of it removed by gauge symmetry?

Paper IV reaches pure Weyl gravity by following the locally reduced Einstein–Weyl phase space to the conformal boundary. The massive and massless helicity-2 branches merge into Jordan systems, the vector modes remain second order, and the scalar direction becomes null under the newly emergent Weyl symmetry. This is a statement about the local linearized solution space after the ordinary $\text{Diff} \times \text{Weyl}$ gauge reduction.

Paper VII [31] asks a different question at the same boundary. It places pure Weyl gravity on the conformal cylinder and treats the fifteen residual conformal transformations of the background as gauge transformations. Which classes survive this additional, finite-dimensional BRST reduction?

This distinction is essential. A mode that appears in a local oscillator decomposition need not survive as a class after the residual charges and their ghosts are imposed. Conversely, the

surviving cohomology need not describe one-particle states: it may instead describe scalar vertices, deformation directions, or integrated observables. Paper VII therefore does not propose another positive quantization of the local conformal graviton. It determines the cohomology of a different quotient.

Under the residual gauge prescription studied in the paper, the complete one-particle sector disappears from the relevant cohomology. Two composite curvature classes survive, one dynamical and one topological. The completed classical statement is the following.

Theorem 20.1 (Classical energy-mode pure-Weyl BV–BFV theorem). *In the quadratic, D -finite, $SO(4)$ -finite cylinder category, choose the closed-universe BFV problem in which all fifteen residual conformal generators are constraints and choose the positive-frequency Lagrangian polarization. Then the complete gauge-fixed classical state complex has*

$$\mathcal{H}_{\text{state}} = \text{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-) \otimes \Lambda^\bullet \mathfrak{so}(4, 2)^*, \quad (49)$$

its centered one-particle cohomology vanishes, and

$$\boxed{H_{\text{res}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2.} \quad (50)$$

The pairing in (50) is induced from the selected field-theoretic BV–BFV and positive-frequency data. Moreover, the normalized $E/A/L$ module completes to an infinite-index Krein space $\mathcal{H}_1$, and

$$\mathcal{K} = \Gamma_s(\mathcal{H}_1) \hat{\otimes} \Lambda^\bullet \mathfrak{so}(4, 2)^* \quad (51)$$

carries a closed densely defined residual differential $\overline{Q}$ with the algebraic state space as a graph core. A bounded Cartan contraction localizes completed cohomology to the unchanged finite centered block. Thus (50) also holds for $H^4(\mathcal{K}, \overline{Q})$. At the reduced Lorentzian field level, the tensor/vector curl factors are normally hyperbolic and the field-induced Cauchy space is

$$(H_{\text{TT}}^1 \oplus L_{\text{TT}}^2)_E \oplus (H_{\text{T}}^{3/2} \oplus H_{\text{T}}^{1/2})_A \oplus (H_{\text{TT}}^1 \oplus L_{\text{TT}}^2)_L. \quad (52)$$

Its positive-frequency harmonic transform is Krein-unitary onto $\mathcal{H}_1$.

The two classes in Theorem 20.1 are deformation or vertex classes, not two graviton states. The theorem is classical, quadratic, energy-mode based, and boundary-choice dependent. It does not establish the curved all-degree auxiliary Green witness, its causal homotopies, the complete covariant/Cauchy pairing comparison, a distributional or Hadamard completion, a positive graviton Hilbert space, nonlinear equivalence, quantum anomaly freedom, or interacting unitarity. A direct same-bundle Bach factorization is an optional strengthening.

20.1 Why study the theory on a cylinder?

Paper VII places the free theory on

$$M = \mathbb{R} \times S^3. \quad (53)$$

Time runs along $\mathbb{R}$, while every spatial slice is a three-sphere. The cylinder is not introduced as a cosmological model; it is a representation of conformal physics in which both the symmetry and the spectrum become especially transparent.

Conceptually, cylinder time translation is generated by the same operator D that measures conformal weight in the corresponding flat-space description. States of definite cylinder energy

therefore correspond to local operators of definite conformal dimension. Technically, compactness of S^3 replaces the continuum of spatial momenta by a discrete harmonic spectrum. The complete coefficient space can then be organized by energy, $SU(2)_L \times SU(2)_R$ quantum numbers, chirality, and particle number.

For specialists, this is the standard radial-quantization relation between a conformal field theory and the cylinder. It makes the $SO(4,2)$ action a graded representation problem and supplies the compact generator used in the homotopy below.

The cylinder also exposes the paper's principal physical choice. Its background has fifteen residual conformal symmetries. Paper VII treats all of them, including D , as residual gauge transformations. If D is instead retained as a physical Hamiltonian or boundary charge, its ghost cannot be used to contract the complex and the result below is not the cohomology of that alternative physical problem.

20.2 Three levels of reduction

It is useful to distinguish three spaces. The linearized equations first define a solution space. Local coordinate and Weyl transformations then identify gauge-equivalent solutions. On the cylinder, Paper VII performs one further quotient by the residual conformal transformations:

$$\text{linearized solutions} \longrightarrow \frac{\text{solutions}}{\text{local Diff} \times \text{Weyl}} \longrightarrow \frac{\text{locally reduced theory}}{\text{residual } SO(4,2)}. \quad (54)$$

Paper IV primarily concerns the middle object; Paper VII concerns the last. A mode can therefore occur in the Paper-IV oscillator classification and still fail to define a Paper-VII class. The two statements concern different quotients and do not contradict one another.

20.3 The locally reduced Weyl module

After local reduction, the one-particle coefficient space is taken to be the parity-complete on-shell Weyl-curvature module

$$\mathcal{W}_+ \oplus \mathcal{W}_-. \quad (55)$$

The Weyl curvature is a natural variable because it is invariant under the local linearized $\text{Diff} \times \text{Weyl}$ transformations. It packages the local gauge-invariant content more directly than the metric perturbation.

Each chirality contains three cylinder towers, denoted E , A , and L . Their invariant form is nondegenerate but indefinite:

$$J_{\text{conf}} = +I_E \oplus (-I_A) \oplus (-I_L). \quad (56)$$

Proper conformal transformations mix these towers, so their signs cannot be assigned independently as if the towers were separate conformal representations.

The full coefficient module for the residual calculation is the bosonic Fock space

$$\mathcal{F}_{\mathcal{W}} = \text{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-), \quad (57)$$

with the multiplicative lift of the one-particle form. Thus the starting point remains an indefinite local oscillator module. Any positivity found afterwards must arise from the cohomological reduction rather than from a re-signing of the local modes.

There is now an all-level geometric theorem for the smooth one-particle quotient. In four dimensions the conformal deformation sequence is a flat BGG complex. On the simply connected conformal cylinder it is a fine resolution of the constant adjoint-tractor system, so

$$H_{\text{def}}^q(\mathbb{R} \times S^3) \cong H^q(S^3; \mathbb{C}) \otimes \mathfrak{so}(4, 2)_{\mathbb{C}}. \quad (58)$$

The metric and curvature slots are therefore globally exact. Combining this fact with $B_{\text{lin}} = C_1^\sharp C_1$ gives the smooth equivariant isomorphism

$$\frac{\ker B_{\text{lin}}}{\text{im } K} \cong \ker C_1^\sharp \cap \ker(C_1^\sharp \star) \cong \mathcal{W}_{\text{geom},+}^{\text{sm}} \oplus \mathcal{W}_{\text{geom},-}^{\text{sm}}. \quad (59)$$

For each E_n, A_n, L_n highest weight, an exact symbolic cylinder calculation now constructs a normalized metric representative, evaluates all components of $C_1 h$, verifies chirality and $C_1^\sharp C_1 h = 0$, and exhibits a nonzero curvature pivot. Naturality, multiplicity one, parity, and equality of the all-level character extend this result to every magnetic component and identify the algebraic positive-energy part with $\mathcal{W}_+ \oplus \mathcal{W}_-$ [10, 11].

For a general reader, this says that every smooth gauge-inequivalent Bach-flat metric perturbation is represented faithfully by a geometric on-shell Weyl curvature. For a specialist, it closes the global smooth local-kernel theorem without an elliptic argument. The explicit cylinder representatives close the finite-energy preimage issue without introducing secular time factors or infinite harmonic expansions. This does not construct the BV/BFV zero-mode split or prove continuity on a chosen Hilbert or Krein completion.

20.4 What BRST cohomology asks

BRST theory is a systematic way to remove gauge redundancy. One introduces an auxiliary ghost variable for every gauge generator and constructs a differential d satisfying

$$d^2 = 0. \quad (60)$$

A closed expression, $d\Psi = 0$, is compatible with the gauge constraints; an exact expression, $\Psi = d\Lambda$, is itself gauge-trivial. What remains is the cohomology

$$H(d) = \frac{\ker d}{\text{im } d}. \quad (61)$$

The word “ghost” has two different meanings here. The BRST ghosts are algebraic variables encoding gauge transformations. They are not the negative-sign propagating modes also called ghosts in higher-derivative gravity.

For the residual algebra $\mathfrak{g} = \mathfrak{so}(4, 2)$ and a coefficient module $V \subset \mathcal{F}_{\mathcal{W}}$, the absolute residual complex is

$$C^q(V) = V \otimes \Lambda^q \mathfrak{g}^*, \quad (62)$$

with the Chevalley–Eilenberg differential. For experts, the result below is an absolute Lie-algebra cohomology computation in the centered cylinder polarization. It includes both the kernel and the incoming image; it is not merely a primary-state count, a character identity, or a search for selected invariant vectors.

20.5 Why an infinite field problem reduces to a finite calculation

A field theory contains arbitrarily high energies and arbitrarily many particles, so an exhaustive Fock-space cohomology calculation initially looks infinite. In this problem a standard homotopy identity reduces it exactly to three finite sectors.

The residual algebra is

$$\mathfrak{g} = \mathfrak{so}(4, 2) = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}, \quad \dim(\mathfrak{g}_{-1}, \mathfrak{g}_0, \mathfrak{g}_{+1}) = (4, 7, 4), \quad (63)$$

where compact time translation D and spatial rotations belong to $\mathfrak{g}_0$, and the proper conformal raising and lowering generators belong to $\mathfrak{g}_{\pm 1}$.

The total compact degree of the complex is the sum of matter and ghost degrees.

The central simplification is the Cartan homotopy identity

$$d\iota_D + \iota_D d = \mathcal{L}_D. \quad (64)$$

Here ι_D contracts with the ghost associated with D , while $\mathcal{L}_D$ measures total compact degree. On every subspace with nonzero degree δ , the operator

$$h_\delta = \frac{\iota_D}{\delta} \quad (65)$$

obeys

$$dh_\delta + h_\delta d = 1. \quad (66)$$

Every closed cochain of nonzero total degree is therefore exact. Only the degree-zero subcomplex can contribute.

This statement requires D to be treated as a residual gauge generator. If cylinder time translation is instead retained as a physical Hamiltonian or boundary charge, contraction by its ghost is not an allowed physical quotient and the homotopy reduction does not apply.

The centered residual ghost representative is the product of the four ghosts dual to the proper conformal raising generators. It has ghost number four and compact degree -4 . At total degree zero, the matter contribution therefore has weight at most four. Since every nonvacuum Weyl quantum has weight at least two, only the sectors

$$N = 0, \quad N = 1, \quad N = 2 \quad (67)$$

can contribute. Every sector with three or more Weyl quanta is eliminated before any large rank calculation is required.

The Cartan identity itself is standard. The nontrivial use made here is that, together with the centered ghost-degree bound, it makes the residual calculation exhaustive rather than merely truncating it for convenience.

20.6 The exact residual theorem

The answer can first be stated without equations: the vacuum contributes no class, no individual one-particle Weyl excitation survives, and exactly two composite curvature classes remain.

Mathematically, the three allowed sectors can be computed exhaustively. For the vacuum coefficient module, ordinary semisimple Lie-algebra cohomology gives no class at the selected centered degree:

$$H_{\delta=0, N=0}^4 = 0. \quad (68)$$

For the complete one-particle Weyl module, an exact absolute kernel-and-image calculation gives

$$H_{\delta=0, N=1}^4 = 0. \quad (69)$$

The exact ranks are

$$\text{rank } d_3 = 260, \quad \text{rank } d_4 = 1051, \quad 260 + 1051 = 1311, \quad (70)$$

which saturates the centered one-particle middle space. Thus none of the indefinite one-particle cylinder modes survives as a class in this residual complex.

The lowest two-particle sector begins at matter weight four. It has no incoming centered ghost-number-three space, and its exact kernel is two-dimensional; equivalently, the outgoing differential has rank 53 on a 55-dimensional middle space. The complete result is

$$H_{\text{res}, \min}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2. \quad (71)$$

The two generators are the chiral scalar primaries formed from two positive- or two negative-chirality Weyl curvatures. Their normalized complementary-degree residual CE pairing is positive.

The ghost insertion entering this pairing is not an arbitrary matrix. The compact grading has dimensions $(4, 7, 4)$: the ket supplies the determinant of the four lowering-degree ghosts, the bra supplies its conjugate, and a middle determinant insertion saturates the seven degree-zero ghosts. Their wedge is the top form on all fifteen ghost directions. Semisimplicity makes the residual algebra unimodular, so the CE differential is graded skew for this complementary-degree pairing and the form descends to residual cohomology. This proves $G_{\text{res}} = I_2$ inside the selected residual complex. The completed BV–BFV comparison below proves that the selected field-theoretic polarization induces this same orientation and normalization.

For a reader who skips the rank calculation, the physical content is:

After the residual conformal gauge reduction, no individual one-particle graviton class remains. Two composite curvature expressions survive, and their normalized residual CE pairing is positive.

The scope is narrower than a positive-particle theorem. The local one-particle module remains indefinite; it has disappeared rather than been made positive. Moreover, G_{res} is unconditionally the normalized field-induced cohomological pairing in the specified algebraic closed-cylinder polarization. This remains distinct from a completed positive-particle norm.

20.7 From the two classes to familiar gravitational terms

The two survivors are not new particle species. They are best understood as directions in the space of local vertices or actions. The four-ghost factor has exactly the compact degree needed to pair with a scalar primary of conformal weight four, and residual descent maps the chiral classes to integrated curvature densities.

Passing to the parity basis,

$$e = \frac{W_+^2 + W_-^2}{\sqrt{2}}, \quad o = \frac{W_+^2 - W_-^2}{\sqrt{2}}, \quad (72)$$

one obtains, up to the conventional Lorentzian self-duality phase,

$$e \longleftrightarrow C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}, \quad o \longleftrightarrow C_{\mu\nu\rho\sigma} \tilde{C}^{\mu\nu\rho\sigma}. \quad (73)$$

The parity-even class varies to the Bach equation and therefore represents the local Weyl-gravity deformation direction. The parity-odd class is the Pontryagin direction. Its variation is a boundary term, so it is trivial in the local bulk Euler–Lagrange quotient, although it may retain global information through boundaries, topology, large transformations, or a gravitational theta parameter.

Consequently,

$$H_{\text{dyn}}^4 \cong \text{span}\{[C^2]\}, \quad G_{\text{dyn,rep}} = I_1, \quad (74)$$

and the representative matrix decomposes as

$$I_2 = I_1^{\text{dynamical}} \oplus I_1^{\text{topological}}. \quad (75)$$

Thus the exact positive I_2 separates into one local dynamical direction and one topological or boundary direction. It does not count two propagating degrees of freedom.

Relation to earlier residual BRST work

The conformal-cylinder residual algebra, the four-ghost centered polarization, and the selection of scalar matter operators of weight four build on the work of Hamada, Horata, and related conformal-cylinder BRST constructions [8, 9]. Those works also exhibited Weyl-square-type physical operators in a broader model containing the conformal or Riegert sector.

The new result of Paper VII is not the isolated observation that a weight-four Weyl-square state can be written. It is the parity-complete pure-Weyl calculation of the full minimal absolute residual window: the Cartan homotopy makes the calculation exhaustive, the vacuum and one-particle sectors are proved acyclic, exactly two two-particle classes survive, and their residual CE pairing and dynamical/topological decomposition are determined.

20.8 From bulk BV fields to residual states

Earlier versions of the programme retained a field-theoretic identification hypothesis between the certified polynomial detour model and the complete gauge-fixed algebraic BV–BFV domain. That historical gap is now closed in the selected classical category. The construction proceeds in six explicit stages.

Minimal BV origin. The starting variables are the unreduced pure-Weyl BV fields

$$(h_{\mu\nu}, c^\mu, \omega; h^{*\mu\nu}, c_\mu^*, \omega^*). \quad (76)$$

Their quadratic tangent differential is derived from the strict pure-Weyl master action before raw polynomial or oscillator coordinates are introduced.

Exact raw-chain identification. An invertible triangular trace/Weyl transformation intertwines the tangent differential with the certified detour chain and gives

$$\boxed{\mathcal{C}_{\text{BV}}^{\text{min,alg}} \cong \mathcal{C}_{\text{raw}}^{\text{tf,min}} \oplus \mathcal{C}_{\text{tr}}, \quad H(\mathcal{C}_{\text{tr}}) = 0.} \quad (77)$$

The trace summand has an explicit homotopy, and the transformation preserves

$$\ker K \cong \mathfrak{so}(4, 2). \quad (78)$$

Thus no residual generator is absorbed into the trace contraction.

Gauge fixing. A specified conformal-Landau gauge fermion, its antighosts, multipliers, and antifield duals define a canonical gauge-fixed extension. Exact projection, inclusion, and homotopy maps contract it back to (77) while preserving all fifteen zero modes. Gauge fixing is therefore an explicitly equivalent realization, not a new assumption.

Dual endpoint and Taub obstruction. For the minimal chain $G \xrightarrow{A} M \xrightarrow{B} E \xrightarrow{C} I$, the perfect endpoint pairing gives the canonical isomorphism

$$\boxed{I/\text{im } C \cong (\ker A)^*}. \quad (79)$$

The bulk endpoint class, the BFV ghost momentum, and the moment-map value are distinct objects with different degrees. The quadratic endpoint obstruction maps under (79) to the conformal Taub/moment map.

BV-to-BFV suspension. The endpoint coordinates are normalized by the duality map Θ . With $\Omega_{\text{gh}} = \delta b_a \wedge \delta c^a$ and the canonical residual BFV charge, compatibility of the bulk and BFV cohomological vector fields fixes

$$\boxed{\tau = \Theta, \quad \lambda = +1.} \quad (80)$$

The local zero-mode cotangent pair is transferred into one residual BFV system, not copied. The residual ghosts c^a , their momenta b_a , and the constraints μ_a retain their separate roles [13].

Positive-frequency polarization. The complexified physical phase space splits into complementary Lagrangians $L_+ \oplus L_-$ with $L_+ = \mathcal{W}_+ \oplus \mathcal{W}_-$. Negative frequencies and antifields supply the symplectic dual rather than independent ket generators; local and nonminimal quartets contribute only their vacuum. The residual b_a act as contractions ι_a on one exterior ghost algebra. This gives (49), the strict residual CE differential, and the field-induced pairing in (50).

The full five-term BGG deformation complex and the four-row tangent BV detour chain remain distinct constructions; endpoint duality is proved directly from the latter rather than inferred by naming its quotient a BGG H^3 representative. Dneprov and Grigoriev's nonlinear Weyl-jet model remains useful comparison data, but its curvature-dependent ghost terms are not inserted into the free residual differential [12]. The reduced metric-field Cauchy realization is now proved; the remaining covariant item is the all-degree local BV Green's witness and its support-sensitive cohomology. Nonlinear transfer and quantum consistency are likewise not consequences of the energy-mode theorem.

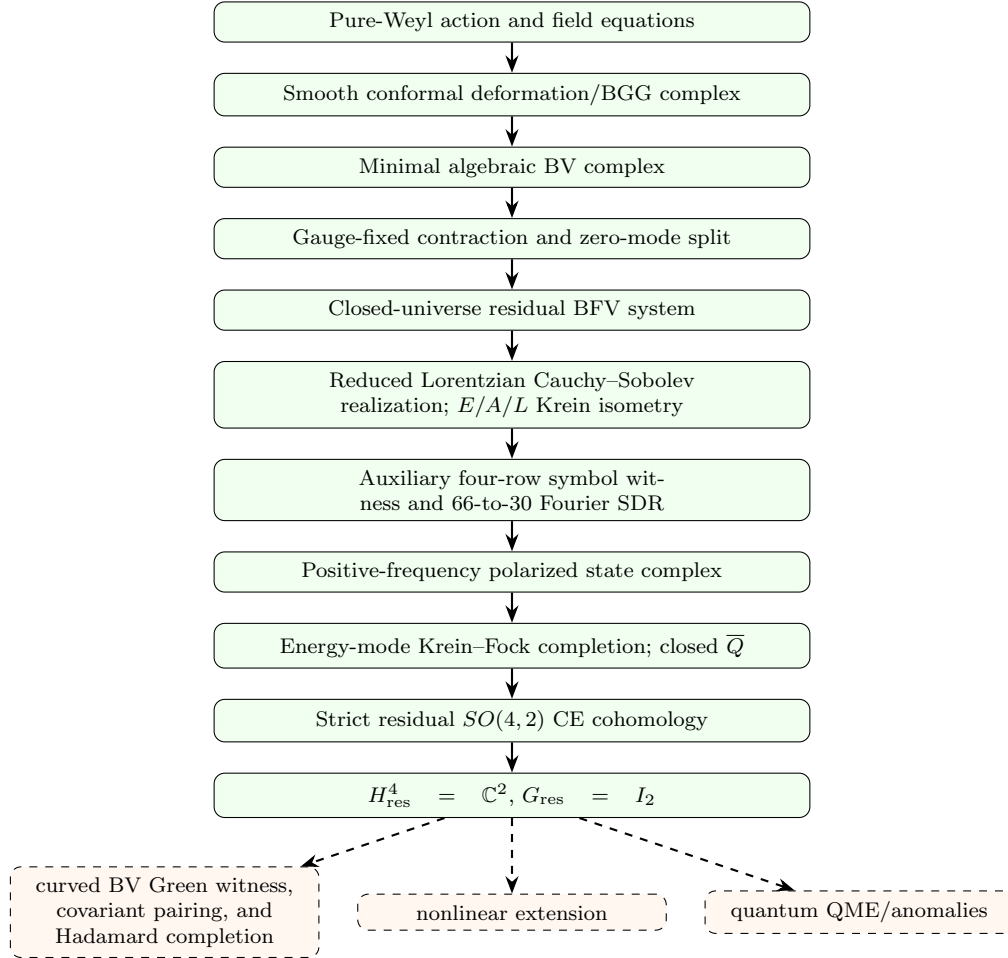


Figure 3: Completed classical theorem chain for the pure-Weyl branch. Solid arrows are proved in the stated smooth, algebraic, or energy-mode categories; dashed arrows mark the three remaining frontiers.

20.9 Why the result matters

For a general reader, the lesson is that the signs of modes in a local oscillator decomposition are not yet the final physical verdict in a gauge theory. One must first determine which objects survive all of the gauge constraints being imposed.

For a specialist, the result cleanly separates three commonly conflated objects:

$$\text{local on-shell module}, \quad \text{absolute residual cohomology}, \quad \text{local BV cohomology}. \quad (81)$$

Paper VII computes the residual object and derives its selected algebraic BV–BFV state complex from the local metric theory. The smooth local BRST group modulo all admissible space-time boundary functionals remains a distinct descent question, but there is no unidentified algebraic model between the metric fields and the centered residual calculation.

The proof hierarchy has three nested levels:

$$\begin{aligned} \frac{\ker B_{\text{lin}}}{\text{im } K} &\cong \mathcal{W}_+ \oplus \mathcal{W}_- && \text{(geometric and algebraic cylinder theorem),} \\ \mathcal{H}_{\text{state}} &= \text{Sym}(\mathcal{W}_+ \oplus \mathcal{W}_-) \otimes \Lambda^\bullet \mathfrak{so}(4, 2)^* && \text{(classical BV–BFV state theorem),} \\ H_{\text{res}}^4 &= \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2 && \text{(centered residual theorem).} \end{aligned} \quad (82)$$

The first identifies available local curvature modes, the second identifies the polarized state coefficient complex, and the third computes the selected cohomology and its field-induced form.

Within the research programme, Paper VII adds a second branch from Paper IV. Papers I–IV classify the local free complex dynamics and its possible real-form completions. Papers V–VI test whether those completions survive physical interactions. Paper VII asks whether the corresponding local modes survive a stronger conformal gauge reduction in the first place.

The result shows that three questions must be kept separate:

$$\begin{aligned} &\text{local oscillator content} \neq \text{residual BRST cohomology}, \\ &\text{residual BRST cohomology} \neq \text{interacting probability structure}. \end{aligned} \quad (83)$$

A locally indefinite one-particle module can have no centered residual one-particle classes, while positive representative matrices survive on scalar vertex directions. This does not solve the interacting ghost problem. It identifies more precisely the objects on which that problem should be posed.

The local equations determine which modes are available. Gauge cohomology determines which states and vertices are independent. Interactions determine whether the surviving structures can support a consistent dynamics.

Paper VII therefore changes the conformal-gravity question from “Can the negative-sign graviton be made positive?” to the more basic question “What is the object on which a positivity question should be asked?”

21 Paper VI: interaction obstruction in Einstein–Weyl gravity

Paper VI [30] asks the interacting question left open by the free gravity classification. Scalar-free Einstein–Weyl gravity around flat space contains a healthy massless graviton h and a

ghost-signed massive spin-2 multiplet M . Paper IV found two covariant free real-form classes: a positive completion obtained by uniformly quarter-turning the complete massive multiplet, and a standard-reality Krein completion with uniformly negative massive signature. Paper V supplied the diagnostic. If ρ_0 is the free Dyson map, an analytic interacting deformation must solve cohomological equations

$$[h_0, R_n] = S_n, \quad (84)$$

so the projection of S_n onto exact free-energy degeneracies must vanish.

Gravity is protected at first order for two independent reasons. The Einstein solution space is an exact nonlinear subsector of Einstein–Weyl gravity, so tree amplitudes with exactly one massive external field and otherwise massless gravitons vanish. The remaining odd-massive cubic vertex MMM is nonzero on complex kinematics but has no real equal-mass $1 \leftrightarrow 2$ shell. Consequently the first-order positive-metric obstruction vanishes.

At second order the process

$$M + M \longrightarrow M + h \quad (85)$$

is physically open. Paper VI establishes its nonvanishing by two independent calculations. An Einstein-frame factorization calculation produces a nonzero massive-pole residue. A separate calculation in the original fourth-order variables includes the quartic contact term, all s -, t -, and u -channel exchanges, physical polarization and equation-of-motion conditions, the reverse-process adjoint, Ward identities, and two internal gauge representatives. At an interior rational point of the real physical shell it gives a nonzero exact amplitude.

The essential bridge is not inferred from that amplitude’s phase alone. In a regulated stationary Fock space, define the second Born operator

$$B_{2,+}(E) = P_E(v_2 + v_1 Q_E(E - h_0)^{-1} Q_E v_1) P_E. \quad (86)$$

With the spectral solution for R_1 , Paper VI proves the identity

$$P_E \left[(v_2^\dagger - v_2) + \frac{1}{2} [R_1, v_1 + v_1^\dagger] \right] P_E = B_{2,+}(E)^\dagger - B_{2,+}(E). \quad (87)$$

The covariant contact-plus-exchange calculation matches this stationary Born operator after the usual propagator decomposition, external residue normalization, and identical-particle factors. The calculation therefore evaluates the complete deformation source, not merely a phase-sensitive scattering amplitude.

Covariance quarter-turns the complete massive multiplet. Counting the massive fields at all vertices and the compensating signs of internal massive propagators shows that every connected diagram with E_M external massive legs acquires the total phase $(-i)^{E_M}$. The process $MM \rightarrow Mh$ has $E_M = 3$; its nonzero real standard-frame Born block thus becomes anti-Hermitian in the positive frame. Equation (87) then gives a nonzero second-order shell cocycle. Hence no analytic second-order deformation based at the canonical Paper-IV positive real form exists in the stated tree-level perturbative class on a nonempty open subset of this shell.

The Krein result is different. Agreement with the free gravitational signature and natural particle-number-diagonal, cluster-multiplicative second quantization gives the conventional lift

$$J_F = (-1)^{N_M} \quad (88)$$

on physical BRST cohomology. The process crosses its positive and negative sectors, but oddness under a $\mathbb{Z}_2$ grading does not imply nullity: the obstruction has a nonzero quadratic Krein trace. Moreover, the verified nonzero MMM and MMh vertices force every uniform abelian charge on the irreducible massive multiplet, with neutral graviton, to be trivial. The scalar boundary's one-sided $O(1,1)$ charge mechanism therefore has no direct gravitational analogue.

This conclusion is deliberately limited. Paper VI excludes an analytic second-order deformation of the canonical positive free metric and a null interpretation under the conventional multiplicative massive-number grading. It does not exclude indefinite Krein evolution, nonanalytic or differently pointed positive metrics, momentum-dependent or non-factorizing involutions, singular conformal-boundary limits, or boundary conditions that truncate part of the fourth-order solution space.

22 What was already solved, and what is new

It is important not to claim that earlier work failed to solve anything. Bender and Mannheim solved the free unequal-frequency oscillator's spectral and positive-norm problem by introducing a nontrivial metric [5, 6]. Bateman and Turok supplied a controlled Krein-space treatment of the massless resonant scalar theory, including a ghost-parity interpretation [7]. The new contribution is a classification and unification (Figure 4):

Paper I [25] reconstructs the Bender–Mannheim generator from the symplectic normal-form data and classifies the diagonalizers and the algebraic metric freedom.

Paper II [26] proves that the canonical Bender–Mannheim transformation is the minimum-Cartan-distortion representative, embeds the result in a general orthogonal projection principle, and separates the physical Dyson equivalence from the auxiliary identity-embedding obstruction.

Paper III [27] separates metric geometry, vacuum covariance, dynamical Jordan structure, and field representation theory; derives the terminating dimension-dependent ultraviolet sector hierarchy; and identifies the common spectral divided-difference covariance, its confluent limit, and its fourth-order Hadamard structure, with the doubly-massless global infrared question left open.

Paper IV [28] determines how gauge reduction and Lorentz covariance change the classification in gravity: the helicity stratification, the exclusion of covariant hybrid completions, covariant projector reassembly and the regular gauge-invariant Weyl correlator, and the conformal Jordan boundary, within the stated free quasifree class.

Paper V [29] tests which free completions survive interactions: the deformation and obstruction theory of Section 15 — generic deformability, resonance obstructions and their transfer lattice (including the computational 5:1 test), spectral complexification in an effective oscillator multiplet, the open-subset continuum shell obstruction, and the exact doubled sector-exchange symmetry.

Paper VI [30] combines the covariant gravity real forms of Paper IV with the deformation complex of Paper V. It proves cubic protection and an external-massive-leg phase theorem, identifies $MM \rightarrow Mh$ as the first open odd-massive channel, derives a nonzero

second-order Born/deformation cocycle from an exact four-point certificate, and shows that the canonical multiplicative massive-number Krein grading does not render the block null. The verified MMM and MMh vertices also forbid a nontrivial uniform abelian charge mechanism analogous to the scalar $O(1, 1)$ boundary.

Paper VII [31] branches from the pure-Weyl boundary of Paper IV. It derives the selected algebraic state complex from the unreduced pure-Weyl BV fields, contracts the trace, gauge-fixed, and nonminimal sectors, transfers the global zero modes into one residual BFV system, and completes the positive-frequency ledger and field pairing. Under full residual conformal gauging the vacuum and one-particle sectors vanish, exactly two weight-four vertex classes survive with $G_{\text{res}} = I_2$, and the local Euler–Lagrange quotient splits them into one dynamical Weyl-square direction and one topological Pontryagin direction.

Completed classical contributions of Paper VII. The introduction synthesizes the following results; their proofs and machine certificates are in Paper VII and its repository snapshot:

1. the global smooth identification of Bach-flat metric perturbations modulo gauge with the parity-complete Weyl-curvature solution module;
2. the exact D -finite $E/A/L$ realization by same-energy metric representatives;
3. derivation of the minimal algebraic BV detour chain from the strict pure-Weyl master action;
4. the trace-sector contraction with preservation of all fifteen conformal-Killing modes;
5. an explicit gauge-fixed and nonminimal contraction;
6. the canonical endpoint duality $I/\text{im } C \cong (\ker A)^*$;
7. identification of the quadratic endpoint obstruction with the conformal Taub/moment map;
8. the normalized zero-mode BV-to-BFV suspension $\lambda = +1$;
9. the complete positive-frequency polarized-state ledger;
10. strict residual $SO(4, 2)$ CE transfer with vanishing relevant higher HPL corrections;
11. the centered theorem $H_{\text{res}}^4 = \text{span}\{[W_+^2], [W_-^2]\}$;
12. the field-induced pairing theorem $G_{\text{res}} = I_2$.

In one paragraph: Papers I–IV classify and geometrically organize the local free theory, culminating in the covariant real forms and conformal boundary of quadratic gravity. The programme then forks. Papers V–VI formulate and execute the interaction-stability test: exactly energy-conserving branch conversion obstructs the canonical positive metric in scalar theory and in the open gravitational channel $MM \rightarrow Mh$, while the conventional gravitational Krein lift does not nullify the latter block. Paper VII returns to the free pure-Weyl boundary

and changes the quotient: after full residual $SO(4, 2)$ gauging, the local indefinite one-particle module has no centered one-particle class, while two positive-representative weight-four vertex directions survive. Local oscillator classification, residual gauge cohomology, and interaction stability are therefore complementary rather than successive versions of one question.

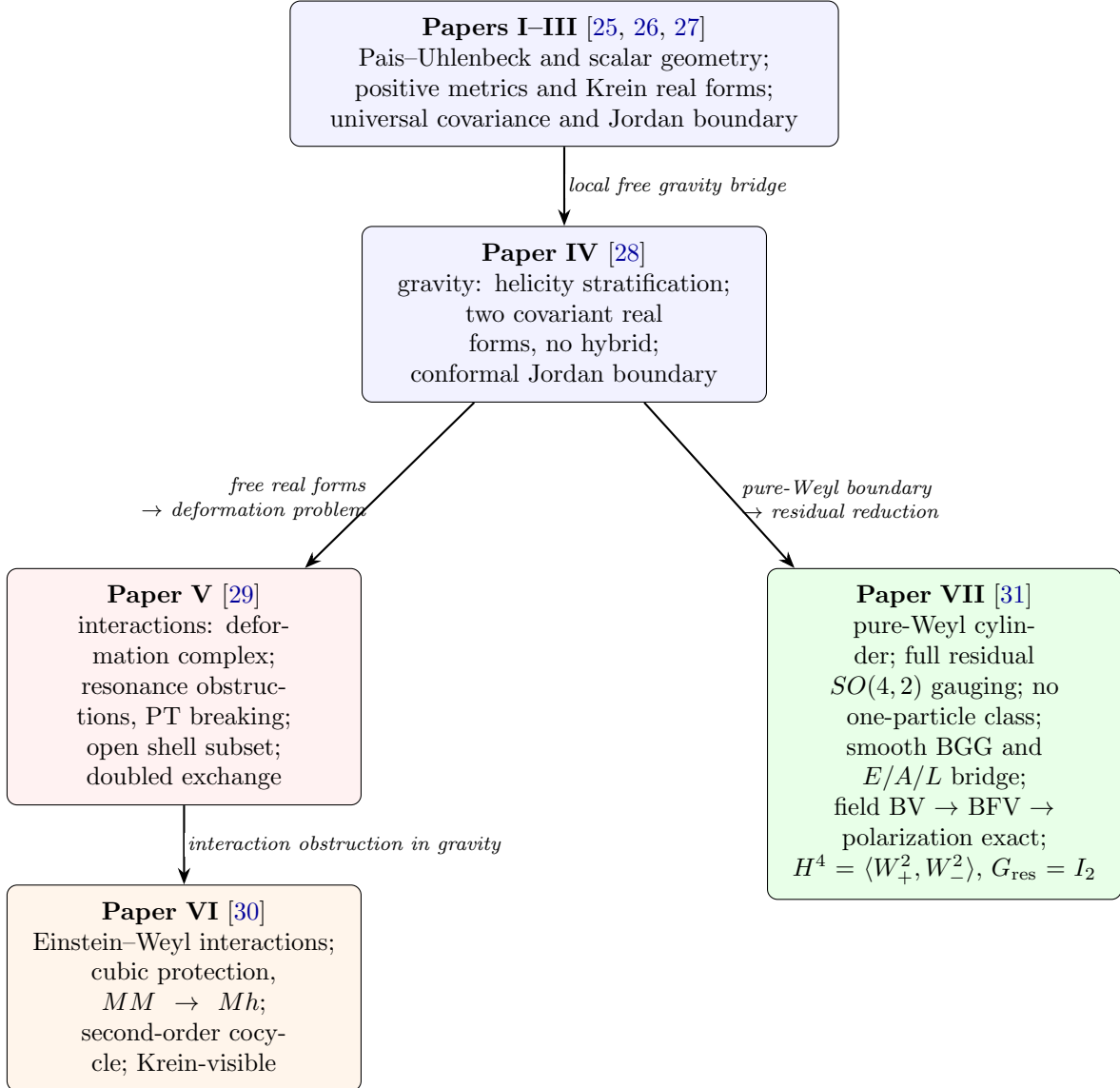


Figure 4: Forked dependency structure of the seven technical papers. Papers I–III establish oscillator and scalar geometry, and Paper IV classifies the local covariant free gravity completions. Papers V–VI form the interaction branch; Paper VII branches from the pure-Weyl boundary to the full residual conformal-gauging problem.

The combined novelty is therefore not a new claim that one particular ghost Hamiltonian can be rewritten positively. It is the statement that:

The known positive and Krein constructions are different real-form completions of a common complex spectral structure. Their local free admissibility is controlled by symplectic geometry, Cartan projection, gauge reduction, Lorentz representation theory, and Jordan degeneration. Residual cohomology determines which classes survive a further gauge quotient; on-shell interactions independently test dynamical continuation.

23 What remains open

The field-theoretic identification and the natural energy-mode Krein–Fock completion of the classical Paper-VII complex are no longer open problems. The frontier now has three parts.

23.1 Covariant analytic completion

The normalized $E/A/L$ modes now complete to an infinite-index Krein space; their bosonic Fock completion carries a closed residual BRST operator, and the bounded Cartan homotopy proves closed range and unchanged centered cohomology. The reduced Lorentzian field comparison is now also exact. Tensor and vector curl give local normally hyperbolic factors for the TT Bach and transverse-vector equations, and the field-action residues give

$$(H_{\text{TT}}^1 \oplus L_{\text{TT}}^2)_E \oplus (H_{\text{T}}^{3/2} \oplus H_{\text{T}}^{1/2})_A \oplus (H_{\text{TT}}^1 \oplus L_{\text{TT}}^2)_L. \quad (89)$$

The harmonic transform from this branch Cauchy space is Krein-unitary onto the energy-mode one-particle completion. The vector order is fixed by the nontrivial residue $R_A = 2(A_A^2 - 4)$; it is not inferred from the second-order equation alone.

The auxiliary route is now exact at symbol and Fourier-retract level. The third-order companion obeys $TK = \square(\square + 2)$ on the full ghost bundle. Rather than assert an unproved direct factorization of the gauge-completed Bach operator, the construction uses Metsaev’s local tensor–tensor–vector formulation [17]. Its exact four-row Fourier symbols obey $Q^2 = 0$, $QW + WQ = P$, and graded symbol-level self-adjointness, with scalar metric principal symbols. The complete 66-component Fourier complex contracts to the 30-component metric BV core. Its Stueckelberg, auxiliary, antifield, and dual maps are finite differential or pointwise algebraic formulas and therefore preserve support. The trace and nonminimal sectors attach as support-local contractible summands.

What remains is to reconstruct the complete curved first- and zeroth-order witness and all-row retract, verify the integration-by-parts adjoints, construct the resulting causal homotopies, and compare the covariant Green current with the fixed-time BV and energy-mode forms, followed by distributional or Hadamard completion. Green-hyperbolic operators and complexes supply the general tools [14, 15]. The direct same-bundle factorization remains an optional strengthening; the known Einstein-background obstruction [18] is not promoted to a no-go theorem for the gauge-completed operator.

Boundary data are part of this problem. The theorem uses the closed-universe constraint choice. A boundary, asymptotic region, external clock, or deparametrization may retain D as a charge and produce a different state complex. Likewise, a Maldacena-type conformal boundary

condition may truncate a bulk branch rather than complete the same bulk space [43]. None of these alternatives is ruled out by the energy-mode theorem.

23.2 Nonlinear classical extension

The next classical question is whether the selected residual constraints are stable under nonlinear evolution. This requires higher Kuranishi and Taub maps, a nonlinear BV-to-BFV transfer, control of possible higher residual operations, and a moduli-space interpretation of the Weyl-square and Pontryagin deformation directions. The strict free CE action need not remain strict after nonlinear transfer merely because it is strict at the quadratic level.

The interaction branch poses complementary questions. Can a nonanalytic or differently pointed positive completion evade the scalar or $MM \rightarrow Mh$ shell cocycles? Can a nonlocal or nonlinear Krein probability structure beyond the conventional massive-number lift remain covariant? How do the answers change with a cosmological constant, scalar modes, matter, or boundary truncations [44, 45]? Does the gravitational cocycle imply complex finite-volume levels or resonance poles, or can an indefinite completion retain a real scattering spectrum?

23.3 Quantum extension

The immediate next phase is the renormalized quantum master equation. It must classify the relevant local anomaly group

$$H^{1,4}(s \mid d), \quad (90)$$

including diffeomorphism and Weyl anomalies, and relate it to the C^2 beta function. One must determine whether the residual constraint algebra and Cartan homotopy survive renormalization, whether the classes W_+^2 and W_-^2 remain quantum BRST classes, and whether their cohomological pairing has a renormalized counterpart.

Only after those questions are answered is it meaningful to ask for a quantum probability interpretation, particle or S -matrix unitarity, or a quantum implementation of the doubled scalar exchange. A recent PT-based proposal for interacting quadratic gravity provides one comparison point [46], but the present classical theorem proves none of these quantum statements.

Do the two classical deformation classes survive an anomaly-free renormalized quantum BV–BFV transfer?

24 Conclusion

The traditional ghost problem combines several logically separate issues: classical stability, spectral boundedness, norm positivity, field reality, vacuum covariance, representation theory, gauge reduction, and Jordan degeneration.

The results now organize into three linked but non-linear questions:

1. Local free classification: identify the complex dynamics, its real forms, canonical metrics, gauge reduction, and Jordan limits.
2. Residual gauge reduction: determine which state, vertex, or observable classes survive when background symmetries are gauged.
3. Interaction stability: test whether a chosen free or reduced structure survives the physical nonlinear processes.

The PU oscillator shows that a Hamiltonian that appears ghostlike in one real form may possess a positive pseudo-Hermitian completion in another; the minimum-distortion theorem explains why the Bender–Mannheim metric is geometrically canonical; the free field theory shows that the metric may become singular while the complex vacuum covariance remains regular; and gravity adds gauge symmetry and Lorentz covariance, which remove some pairings and force the massive spin-2 multiplet to be treated as a whole.

The interaction results extend the picture: the positive pseudo-Hermitian construction is a valid local resolution of the free and, through the computed perturbative orders, nonresonant weakly interacting theory. It is not generically stable under exactly energy-conserving branch-changing processes: in the field calculation the analytic deformation of the canonical pointed metric is obstructed on a verified nonempty open shell subset. Continuous momentum makes the relevant shells kinematically available far more generally than in a finite oscillator. In Einstein–Weyl gravity the exact Einstein subsector protects cubic order, but the physical $MM \rightarrow Mh$ channel yields a second-order cocycle; unlike the scalar boundary obstruction, this block is non-null for the canonical massive-number Krein lift. At the massless perfect-square boundary, an exact exchange between two oppositely oriented Jordan sectors survives in the extended theory, and the regulated charge image becomes one-sided. This boundary nullity mechanism is distinct from the exact Krein pseudo-unitarity already present at the verified split shell.

The residual branch is now complete in its stated classical energy-mode category. It begins with the metric pure-Weyl field theory, derives the minimal and gauge-fixed BV complexes, contracts the trace and nonminimal sectors, transfers the fifteen global zero modes to the selected closed-universe BFV system, and then passes to the positive-frequency state polarization. The resulting strict residual $SO(4, 2)$ complex has no centered one-particle class and has exactly two weight-four curvature-square classes, with field-induced Gram matrix I_2 . One direction is dynamical and one topological. These are vertex or deformation classes, not two positive-norm gravitons. The algebraic state complex has a natural infinite-index Krein–Fock completion with closed residual differential and the same cohomology. The reduced Lorentzian field equations now also have an exact Green factorization and Cauchy–Sobolev realization equivalent to that one-particle completion. This is not yet a complete BV Green-complex, a nonlinear theorem, or a quantum statement.

The seven papers therefore form a fork rather than a single progression. Papers I–IV classify the local free structures. Papers V–VI test their interacting continuation. Paper VII returns to the pure-Weyl boundary and tests an additional residual gauge quotient. No single free-field slogan resolves all three questions. In scalar theories, energy-conserving conversion obstructs the canonical positive metric while an enhanced doubled structure can survive at a special boundary. In Einstein–Weyl gravity, gauge reduction and covariance first restrict the free choices, and $MM \rightarrow Mh$ then obstructs the analytic continuation of the canonical positive one. The conventional gravitational Krein grading remains algebraically available but does not make that cocycle null.

The resulting view is neither that higher-derivative ghosts are automatically harmless nor that they automatically invalidate a theory. The refined question is:

Not “Can the ghost be removed?” but “Which local completion, residual quotient, observable algebra, and probability structure are compatible with the physical problem being posed?”

Once the ingredients — complex dynamics, real form, involution, observable algebra, local and residual quotient, completion, and interaction stability — are separated, the existing positive and indefinite constructions cease to look like unrelated tricks. They become different parts of one geometric, representation-theoretic, and deformation-theoretic framework.

The programme therefore now supplies a complete classical energy-mode BV–BFV derivation of the centered residual cohomology of free pure-Weyl gravity on the conformal cylinder, including its infinite-index Krein–Fock completion and reduced field Cauchy–Sobolev realization. The remaining frontier is not the identification or natural mode completion of the classical complex: it is the curved all-degree Green witness, complete covariant/Cauchy pairing and Hadamard completion, nonlinear extension, and renormalized quantum master equation.

Do the two classical deformation classes survive an anomaly-free renormalized quantum BV–BFV transfer?

Verification. Every finite-dimensional, variational, symplectic, and modewise distributional identity quoted in this introduction is machine-verified in the repository accompanying the series [25, 26, 27, 28, 29, 30, 31] (symbolic SymPy audits, high-precision numerical regression, and a Lean 4 formalization of the paper-I core with zero unproved placeholders); the technical papers state precisely which of their conclusions are analytic rather than machine-checked.

References

- [1] M. Ostrogradsky, *Mémoire sur les équations différentielles relatives au problème des isopérimètres*, Mém. Acad. St. Pétersbourg **VI** 4 (1850) 385.
- [2] R. P. Woodard, *The theorem of Ostrogradsky*, arXiv:1506.02210.
- [3] A. Pais and G. E. Uhlenbeck, *On field theories with non-localized action*, Phys. Rev. **79** (1950) 145.
- [4] C. M. Bender and S. Boettcher, *Real spectra in non-Hermitian Hamiltonians having PT symmetry*, Phys. Rev. Lett. **80** (1998) 5243.
- [5] C. M. Bender and P. D. Mannheim, *No-ghost theorem for the fourth-order derivative Pais–Uhlenbeck oscillator model*, Phys. Rev. Lett. **100** (2008) 110402.
- [6] C. M. Bender and P. D. Mannheim, *Exactly solvable PT-symmetric Hamiltonian having no Hermitian counterpart*, Phys. Rev. D **78** (2008) 025022, arXiv:0804.4190.
- [7] S. Bateman and N. Turok, *Escape from Ostrogradsky via hidden ghost parity*, arXiv:2607.00096.
- [8] K.-j. Hamada and S. Horata, *Conformal algebra and physical states in non-critical 3-brane on $\mathbb{R} \times S^3$* , Prog. Theor. Phys. **110** (2003) 1169–1210, arXiv:hep-th/0307008.
- [9] K.-j. Hamada, *BRST analysis of physical fields and states for 4D quantum gravity on $\mathbb{R} \times S^3$* , Phys. Rev. D **86** (2012) 124006, arXiv:1202.4538.

- [10] A. R. Gover and L. J. Peterson, *The ambient obstruction tensor and the conformal deformation complex*, Pacific J. Math. **226** (2006) 309–351, arXiv:math/0408229.
- [11] A. Čap, *Overdetermined systems, conformal geometry, and the BGG complex*, in *Symmetries and Overdetermined Systems of Partial Differential Equations*, IMA Vol. Math. Appl. **144** (Springer, 2008) 1–24, arXiv:math/0610225.
- [12] I. Dneprov and M. Grigoriev, *Presymplectic BV–AKSZ formulation of conformal gravity*, arXiv:2208.02933.
- [13] A. S. Cattaneo, P. Mnev and N. Reshetikhin, *Classical BV theories on manifolds with boundary*, Commun. Math. Phys. **332** (2014) 535–603, arXiv:1201.0290.
- [14] C. Bär, *Green-hyperbolic operators on globally hyperbolic spacetimes*, Commun. Math. Phys. **333** (2015) 1585–1615, arXiv:1310.0738.
- [15] M. Benini, G. Musante and A. Schenkel, *Green hyperbolic complexes on Lorentzian manifolds*, Commun. Math. Phys. (2023), arXiv:2207.04069.
- [16] M. Beccaria, X. Bekaert and A. A. Tseytlin, *Partition function of free conformal higher spin theory*, JHEP **08** (2014) 113, arXiv:1406.3542.
- [17] R. R. Metsaev, *Ordinary-derivative formulation of conformal low-spin fields*, JHEP **01** (2008) 064, arXiv:0707.4437.
- [18] T. Nutma and M. Taronna, *On conformal higher spin wave operators*, JHEP **06** (2014) 066, arXiv:1404.7452.
- [19] P. E. T. Jorgensen and F. Tian, *Unbounded operators, Lie algebras, and local representations*, J. Math. Phys. **55** (2014) 111702, arXiv:1406.6966.
- [20] F. G. Scholtz, H. B. Geyer and F. J. W. Hahne, *Quasi-Hermitian operators in quantum mechanics and the variational principle*, Ann. Phys. **213** (1992) 74.
- [21] A. Mostafazadeh, *Pseudo-Hermitian representation of quantum mechanics*, Int. J. Geom. Meth. Mod. Phys. **7** (2010) 1191.
- [22] P. D. Mannheim, *Solution to the ghost problem in higher-derivative gravity*, arXiv:2109.12743.
- [23] P. D. Mannheim, *CPT symmetry without Hermiticity*, arXiv:1611.02100.
- [24] K. S. Stelle, *Classical gravity with higher derivatives*, Gen. Rel. Grav. **9** (1978) 353; Phys. Rev. D **16** (1977) 953.
- [25] *Canonical positive symplectic diagonalization of the Pais–Uhlenbeck oscillator*, companion paper (Paper I) and verification repository, 2026.
- [26] *The Pais–Uhlenbeck metric as a minimum-distortion principle, and the representation problem for the fourth-order field*, companion paper (Paper II), 2026.
- [27] *The universal vacuum of the fourth-order scalar field: metric orbits, Fock sectors, and the Krein boundary*, companion paper (Paper III), 2026.

- [28] *Gauge reduction and the completion problem in fourth-order gravity: PU pairing, covariant real forms, and the conformal Jordan boundary*, companion paper (Paper IV), 2026.
- [29] *Interaction obstructions, resonant PT breaking, and doubled Jordan symmetry in fourth-order theories*, companion paper (Paper V), 2026.
- [30] *Interaction obstructions in Einstein–Weyl gravity: cubic protection, second-order metric failure, and Krein visibility*, companion paper (Paper VI), 2026.
- [31] *Residual $SO(4, 2)$ cohomology of free Weyl gravity on the conformal cylinder: Cartan homotopy, Weyl-square classes, and their residual pairing*, companion paper (Paper VII), 2026.
- [32] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.
- [33] R. Bhatia, *Positive Definite Matrices*, Princeton University Press, 2007.
- [34] G. D. Mostow, *Self-adjoint groups*, Ann. of Math. **62** (1955) 44.
- [35] G. Kempf and L. Ness, *The length of vectors in representation spaces*, in: Algebraic Geometry, Lecture Notes in Math. **732**, Springer, 1979, 233.
- [36] I. Gohberg, P. Lancaster and L. Rodman, *Indefinite Linear Algebra and Applications*, Birkhäuser, 2005.
- [37] T. Kato, *Perturbation Theory for Linear Operators*, Springer, 1966.
- [38] W. D. Heiss, *The physics of exceptional points*, J. Phys. A **45** (2012) 444016.
- [39] J. von Neumann, *On infinite direct products*, Compos. Math. **6** (1939) 1.
- [40] M. J. Radzikowski, *Micro-local approach to the Hadamard condition in quantum field theory on curved space-time*, Commun. Math. Phys. **179** (1996) 529.
- [41] R. Brunetti and K. Fredenhagen, *Microlocal analysis and interacting quantum field theories*, Commun. Math. Phys. **208** (2000) 623.
- [42] S. Hollands and R. M. Wald, *Local Wick polynomials and time ordered products of quantum fields in curved spacetime*, Commun. Math. Phys. **223** (2001) 289.
- [43] J. Maldacena, *Einstein gravity from conformal gravity*, arXiv:1105.5632.
- [44] H. Lü and C. N. Pope, *Critical gravity in four dimensions*, Phys. Rev. Lett. **106** (2011) 181302, arXiv:1101.1971.
- [45] S. Deser, E. Joung and A. Waldron, *Partial masslessness and conformal gravity*, J. Phys. A **46** (2013) 214019, arXiv:1208.1307.
- [46] J. Kubo and J. Kuntz, *Unitarity through PT symmetry in quantum quadratic gravity*, arXiv:2410.08278.